Vector Calculus (MA1130)

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Preface

These notes have been prepared for the course **Vector Calculus (MA1130)**, a 1-credit course offered during Segments $5 - 6$ of the Fall Semester 2026 at the Indian Institute of Technology Hyderabad (IITH). They are based on the lectures delivered in the course, and their primary purpose is to provide a concise and organized presentation of the topics prescribed in the course syllabus.

These notes are neither self-contained nor intended to serve as a comprehensive textbook on calculus. There are many excellent textbooks on the subject that offer a more comprehensive and detailed treatment. Students are encouraged to consult these standard references for additional examples, exercises, and deeper insights.

Many of the definitions, examples, exercises, and proofs presented in these notes have been adapted from or inspired by several standard texts, including *Vector Analysis and an Introduction to Tensor Analysis* by Murray R. Spiegel and lecture notes named *Vector Calculus* by Michael Corral, Prof. Sukumar D, Prof. M Rajesh Kannan, among others. I gratefully acknowledge these valuable sources.

The material presented here is intended to complement the classroom lectures by emphasizing the fundamental concepts, definitions, theorems, proofs, and illustrative examples discussed during the course. Every effort has been made to keep the exposition clear, concise, and accessible to first-year B.Tech students.

Despite every effort to ensure accuracy, these notes may still contain typographical errors or other inadvertent mistakes. I would be grateful if readers could bring any such errors or suggestions for improvement to my attention by email.

Finally, I would like to acknowledge the assistance of AI tools in preparing the LaTeX typesetting and formatting of these notes, which significantly reduced the time required for their preparation.

Introduction

Calculus is one of the most important tools in mathematics. In elementary calculus, we study functions of a single variable such as $f(x)$, where the value of the function depends on only one quantity. Many real-world phenomena depend on several variables. For example, the temperature in a room depends on the position of a point and may be represented by $T(x, y, z)$, while the height of a landscape can be described by a function $z = f(x, y)$. Such functions are studied in multivariable calculus.

In many applications, it is not enough to describe only the magnitude of a quantity. We often need both its magnitude and its direction. Quantities of this type are called *vectors*. Examples include velocity, force, acceleration, and electric field. A collection of vectors assigned to every point of a region is called a *vector field*. For instance, the velocity of water in a river or the wind velocity in the atmosphere can be represented by vector fields.

One of the fundamental questions in multivariable calculus is to understand how scalar and vector fields change from one point to another. For a scalar field such as the temperature $T(x, y, z)$, we may ask:

> In which direction does the temperature increase most rapidly?

The mathematical tool that answers this question is the *gradient*. Similarly, for a vector field, we are interested in measuring how the field spreads out from a point or how it rotates around a point. These ideas lead to the notions of *divergence* and *curl*.

Integration also plays an important role in several variables. Double and triple integrals are used to compute areas, volumes, masses, and other physical quantities. In vector calculus, integration is extended further. Instead of integrating over intervals or regions, we may integrate along curves or over surfaces. Such integrals arise naturally in applications. For example, the work done by a force $\mathbf{F}$ in moving an object along a curve C is given by $\int_C \mathbf{F} \cdot d\mathbf{r}$, while the flow of a fluid through a surface S is measured by the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.

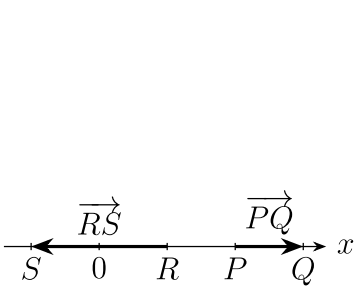
A remarkable feature of vector calculus is that differentiation and integration are closely related. Some of the most important results in mathematics establish deep connections between line integrals, surface integrals, and multiple integrals. These results form the foundation of many applications in mathematics, physics, and engineering.

This course aims to introduce the basic concepts of multivariable and vector calculus, develop techniques for computing multiple and vector integrals, and study the fundamental ideas that connect them. Throughout the course, mathematical concepts will be motivated and illustrated through a variety of examples.

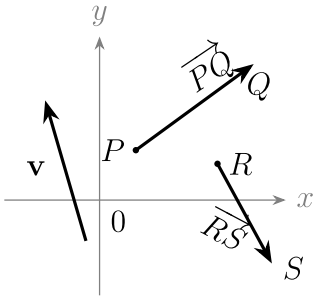
1 Vectors in Euclidean space

1.1 Vector algebra

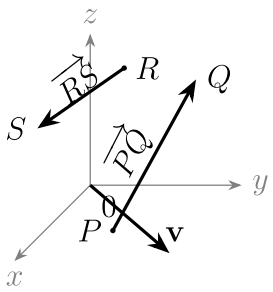
A ***vector*** is a quantity having both magnitude and direction, such as displacement, velocity, force, and acceleration. Graphically, a vector is represented by a directed line segment (or an arrow) $\overrightarrow{PQ}$. The direction of the arrow specifies the direction of the vector, while its length represents the magnitude of the vector. The tail end P of the arrow is called the *origin* or *initial point* of the vector, and the head Q is called the *terminal point* or *terminus*.



(a) One dimension



(b) Two dimensions



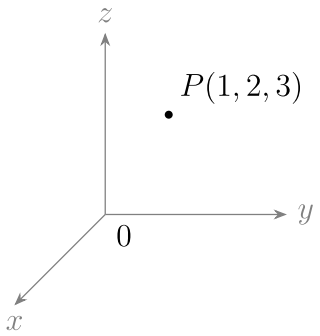
(c) Three dimensions

Definition 1. *A* (nonzero) vector *is a directed line segment from a point P, called the* initial point*, to a distinct point Q, called the* terminal point*. The vector is denoted by $\overrightarrow{PQ}$. Its* magnitude *is the length of the directed line segment and is denoted by $\|\overrightarrow{PQ}\|$. The* direction *of the vector is the direction from P to Q. The* zero vector *has magnitude zero and is denoted by $\mathbf{0}$.*

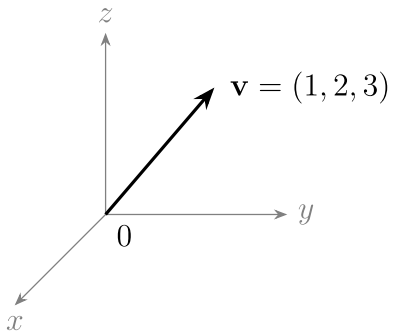
- The zero vector has magnitude 0, that is, $\|\mathbf{0}\| = 0$. Since it has no well-defined direction, the notion of direction is defined only for nonzero vectors.

Definition 2. *Two nonzero vectors are said to be* equal *if they have the same magnitude and the same direction. Any vector with zero magnitude is equal to the zero vector.*

- Unless stated otherwise, a vector will be assumed to have its initial point at the origin of the coordinate system.



(a) The point $(1, 2, 3)$.



(b) The vector $(1, 2, 3)$.

Let $P = (x_1, y_1, z_1)$ and $Q = (x_2, y_2, z_2)$ be two points in $\mathbb{R}^3$. The vector from P to Q is defined by

$$\overrightarrow{PQ} = Q - P = (x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1).$$

The magnitude of $\overrightarrow{PQ}$ is

$$\|\overrightarrow{PQ}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2},$$

which is precisely the distance between the points P and Q. Hence,

$$d(P, Q) = \|\overrightarrow{PQ}\|.$$

A ***scalar*** is a quantity having magnitude but no direction, such as mass, length, time, temperature, and any real number. Scalars are denoted by ordinary letters, and operations involving scalars follow the usual rules of elementary algebra.

Definition 3. *Let k be a scalar and $\mathbf{v}$ be a nonzero vector. The* scalar multiple *of $\mathbf{v}$ by k, denoted by $k\mathbf{v}$, is the vector whose magnitude is $|k|\,\|\mathbf{v}\|$. Moreover,*

- *if $k > 0$, then $k\mathbf{v}$ has the same direction as $\mathbf{v}$;*
- *if $k < 0$, then $k\mathbf{v}$ has the opposite direction to $\mathbf{v}$;*
- *if $k = 0$, then $k\mathbf{v} = \mathbf{0}$.*

For the zero vector, we define $k\mathbf{0} = \mathbf{0}$ for every scalar k.

- The algebra of vectors, including vector addition and subtraction, is assumed to be familiar from the plus two level coordinate geometry and vectors. Readers are encouraged to briefly review these topics before proceeding.

Definition 4. *A vector* $\mathbf{u}$ *is called a* unit vector *if its magnitude is* 1*, that is,*

$$\|\mathbf{u}\| = 1.$$

If $\mathbf{v}$ *is a nonzero vector, then the unit vector in the direction of* $\mathbf{v}$ *is given by*

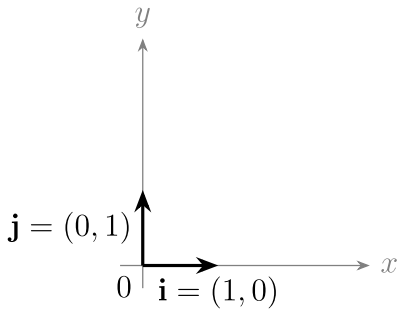
$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

- In $\mathbb{R}^2$, the standard basis vectors are

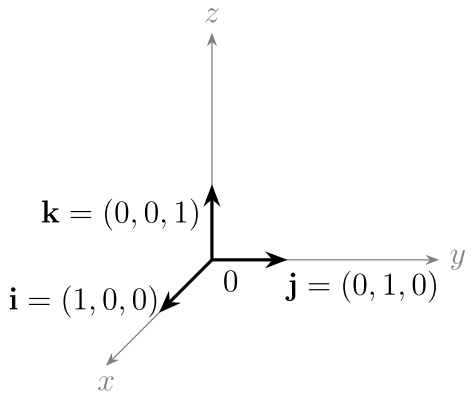
$$\mathbf{i} = (1, 0), \qquad \mathbf{j} = (0, 1),$$

and in $\mathbb{R}^3$, the standard basis vectors are

$$\mathbf{i} = (1, 0, 0), \qquad \mathbf{j} = (0, 1, 0), \qquad \mathbf{k} = (0, 0, 1).$$



(a) Standard basis of $\mathbb{R}^2$.



(b) Standard basis of $\mathbb{R}^3$.

Let $\mathbf{v}$ be a vector in $\mathbb{R}^3$. Then $\mathbf{v}$ can be expressed uniquely as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k},$$

where v_1, v_2, and v_3 are called the *components* of $\mathbf{v}$ along the x, y, and z axes, respectively. Equivalently,

$$\mathbf{v} = (v_1, v_2, v_3).$$

The magnitude of $\mathbf{v}$ is given by

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The *position vector* (or *radius vector*) of a point $P = (x, y, z)$ is the vector from the origin O to P, denoted by

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (x, y, z).$$

Its magnitude is

$$\|\mathbf{r}\| = \sqrt{x^2 + y^2 + z^2}.$$

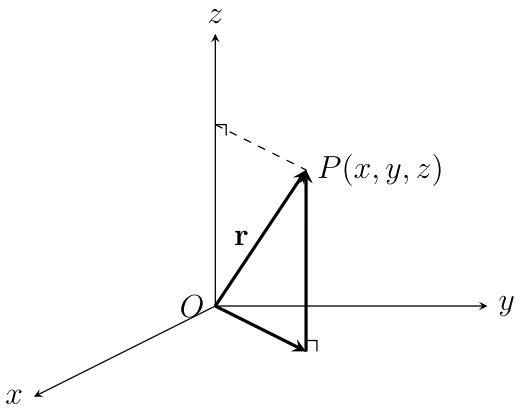
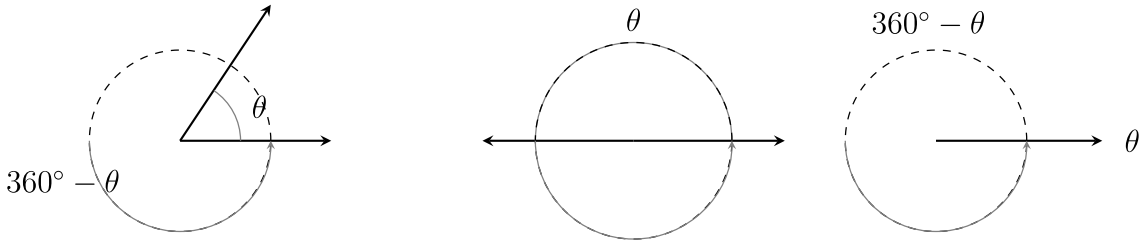


Figure 4: The position vector $\mathbf{r} = \overrightarrow{OP}$ and its components.

1.2 The dot and cross product

The **angle** between two nonzero vectors with the same initial point is the smallest angle between them.



(a) $0° < \theta < 180°$ **(b)** $\theta = 180°$ **(c)** $\theta = 0°$

Figure 5: The angle between two vectors.

The angle between the zero vector and any vector is not defined. For two nonzero vectors with the same initial point, there are two possible angles between them, namely θ and $360° - \theta$. Unless stated otherwise, the *angle between two vectors* will always mean the smaller angle, where

$$0° \leq \theta \leq 180°.$$

Definition 5. *Let* $\mathbf{v} = (v_1, v_2, v_3)$ *and* $\mathbf{w} = (w_1, w_2, w_3)$ *be vectors in* $\mathbb{R}^3$*. The* dot product *(or* scalar product*) of* $\mathbf{v}$ *and* $\mathbf{w}$*, denoted by* $\mathbf{v} \cdot \mathbf{w}$*, is defined by*

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3.$$

Theorem 1. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors, and let* $\theta \in [0, \pi]$ *denote the angle between them. Then*

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \, \|\mathbf{w}\| \cos\theta.$$

In particular, $\mathbf{v}$ *and* $\mathbf{w}$ *are perpendicular if and only if*

$$\mathbf{v} \cdot \mathbf{w} = 0.$$

- Readers are encouraged to briefly review the basic properties of the dot product from the plus two level before proceeding.

Definition 6. *Let* $\mathbf{v} = (v_1, v_2, v_3)$ *and* $\mathbf{w} = (w_1, w_2, w_3)$ *be vectors in* $\mathbb{R}^3$*. The* cross product *(or* vector product*) of* $\mathbf{v}$ *and* $\mathbf{w}$*, denoted by* $\mathbf{v} \times \mathbf{w}$*, is defined by*

$$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2,\ v_3 w_1 - v_1 w_3,\ v_1 w_2 - v_2 w_1).$$

Theorem 2. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors. If* $\mathbf{v} \times \mathbf{w} \neq \mathbf{0}$*, then* $\mathbf{v} \times \mathbf{w}$ *is perpendicular to both* $\mathbf{v}$ *and* $\mathbf{w}$*.*

Theorem 3. *Let* $\mathbf{v}$ *and* $\mathbf{w}$ *be nonzero vectors in* $\mathbb{R}^3$*, and let* θ *denote the angle between them. Then*

$$\|\mathbf{v} \times \mathbf{w}\| = \|\mathbf{v}\| \, \|\mathbf{w}\| \sin\theta.$$

- Readers are encouraged to review the basic properties of the cross product, its relationship with the dot product, and the standard geometric applications of vector algebra, including formulas for the area of a triangle, the area of a parallelogram, and the volume of a parallelepiped, from the plus two level before proceeding.

Theorem 4 (Area of Triangles and Parallelograms)**.** *(a) The area* A *of a triangle*

with adjacent sides $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ *is*

$$A = \frac{1}{2}\|\mathbf{v} \times \mathbf{w}\|.$$

(b) The area A *of a parallelogram with adjacent sides* $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ *is*

$$A = \|\mathbf{v} \times \mathbf{w}\|.$$

1.3 Lines, planes and surfaces

In this section, we study how vectors can be used to represent lines in three-dimensional Euclidean space.

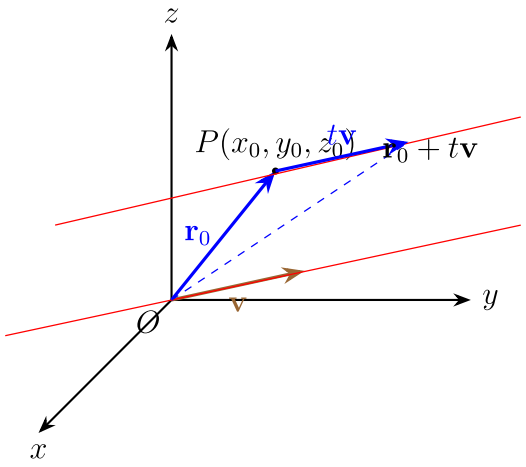


Figure 6: Vector equation of a line.

Let $P = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$\mathbf{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = (a, b, c) \neq \mathbf{0}$$

be a direction vector. Then the line passing through P and parallel to $\mathbf{v}$ can be represented in the following equivalent forms:

(i) **Vector equation:**

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \qquad t \in \mathbb{R},$$

where $\mathbf{r}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ is the position vector of P.

(ii) **Parametric equations:**

$$x = x_0 + at, \qquad y = y_0 + bt, \qquad z = z_0 + ct, \qquad t \in \mathbb{R}.$$

(iii) **Symmetric equations:** If a, b, and c are all nonzero, then

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

- A *parametric representation* of a curve is a representation in which the coordinates of every point on the curve are expressed as functions of a parameter.

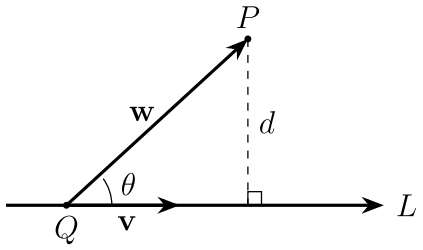


Figure 7: Distance from a point to a line.

Let

$$L: \ \mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \qquad t \in \mathbb{R},$$

be a line in $\mathbb{R}^3$, where $\mathbf{v} \neq \mathbf{0}$, and let P be a point not lying on L. Choose any point Q on L, and let

$$\mathbf{w} = \overrightarrow{QP}.$$

Then the perpendicular distance from P to L is given by

$$d(P, L) = \frac{\|\mathbf{v} \times \mathbf{w}\|}{\|\mathbf{v}\|}.$$

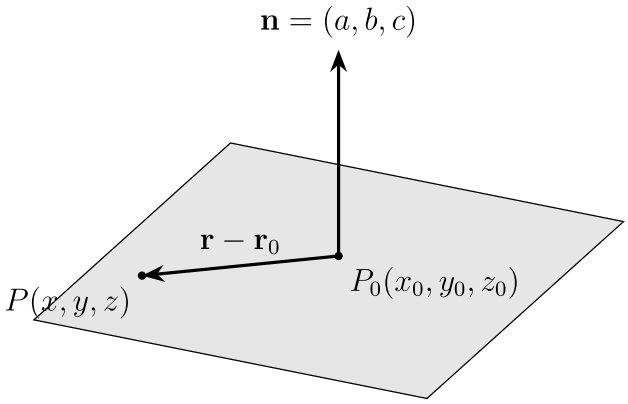


Figure 8: A plane with normal vector $\mathbf{n}$.

Let $P_0 = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$\mathbf{n} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} = (a, b, c) \neq \mathbf{0}$$

be a normal vector to a plane. Then the plane passing through P_0 with normal vector $\mathbf{n}$ can be represented in the following equivalent forms:

(i) **Vector equation:**

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of an arbitrary point (x, y, z) on the plane, and $\mathbf{r}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ is the position vector of P_0.

(ii) **Point-normal form:**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

(iii) **Normal form:**

$$ax + by + cz + d = 0.$$

Let $Q = (x_0, y_0, z_0)$ be a point in $\mathbb{R}^3$, and let

$$ax + by + cz + d = 0$$

be the equation of a plane not containing Q. Then the perpendicular distance from Q to the plane is given by

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

In the previous part, we studied lines and planes in Euclidean space. We now turn our attention to *surfaces* in $\mathbb{R}^3$. Informally, a surface is the set of all points (x, y, z) satisfying an equation of the form

$$F(x, y, z) = 0,$$

where $F : \mathbb{R}^3 \to \mathbb{R}$ is a given function. Thus, the plane

$$ax + by + cz + d = 0$$

is a surface corresponding to the function

$$F(x, y, z) = ax + by + cz + d.$$

In this part, we introduce some of the most common surfaces in $\mathbb{R}^3$ and discuss their equations and graphical representations.

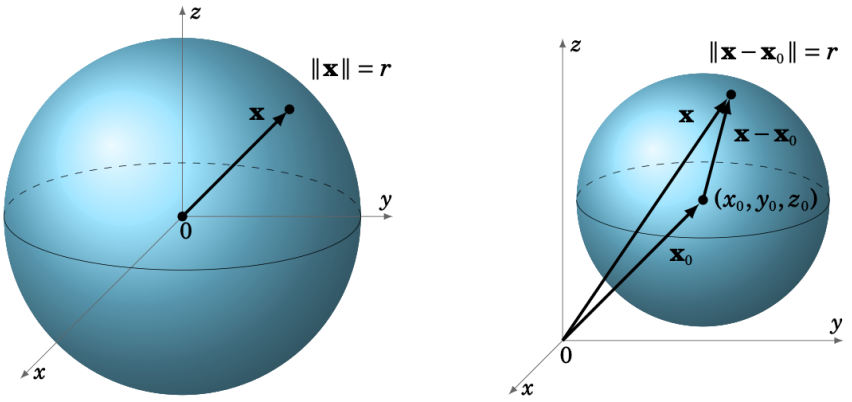
A *sphere* with center $P_0 = (x_0, y_0, z_0)$ and radius $r > 0$ is the set of all points $(x, y, z) \in \mathbb{R}^3$ whose distance from P_0 is equal to r. Thus,

$$S = \left\{(x, y, z) \in \mathbb{R}^3 : (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2\right\}.$$

Equivalently, in vector notation,

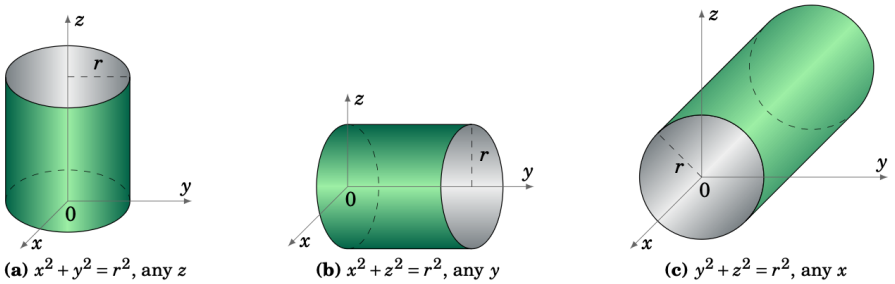
$$S = \left\{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x} - \mathbf{x}_0\| = r\right\},$$

where $\mathbf{x} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $\mathbf{x}_0 = x_0\mathbf{i} + y_0\mathbf{j} + z_0\mathbf{k}$ are the position vectors of (x, y, z) and P_0, respectively.



(a) radius r, center $(0,0,0)$ **(b)** radius r, center (x_0, y_0, z_0)

In this course, we consider only *right circular cylinders*. Such a cylinder is obtained by moving a straight line parallel to itself along a circle in $\mathbb{R}^3$, with the line remaining perpendicular to the plane containing the circle. Throughout this course, we restrict our attention to cylinders whose axes are parallel to one of the coordinate axes.



(a) $x^2 + y^2 = r^2$, any z **(b)** $x^2 + z^2 = r^2$, any y **(c)** $y^2 + z^2 = r^2$, any x

Figure 1.6.3 Cylinders in $\mathbb{R}^3$

An *ellipsoid* is the set

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

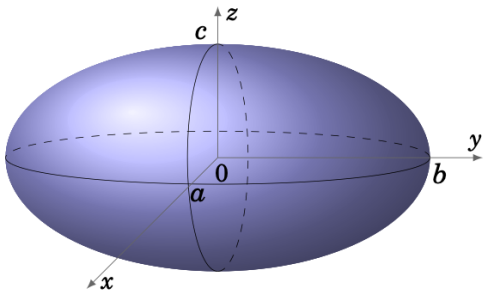


Figure 1.6.4 Ellipsoid

A *hyperboloid of one sheet* is the set

$$H = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

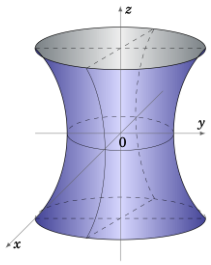


Figure 1.6.5 Hyperboloid of one sheet

A *hyperboloid of two sheets* is the set

$$H = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \right\},$$

where $a, b, c > 0$.

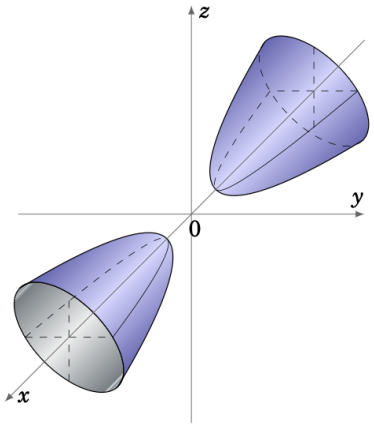


Figure 1.6.6 Hyperboloid of two sheets

An *elliptic paraboloid* is the set

$$P = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} \right\},$$

where $a, b, c > 0$. If $c < 0$, the paraboloid simply opens downward instead of upward.

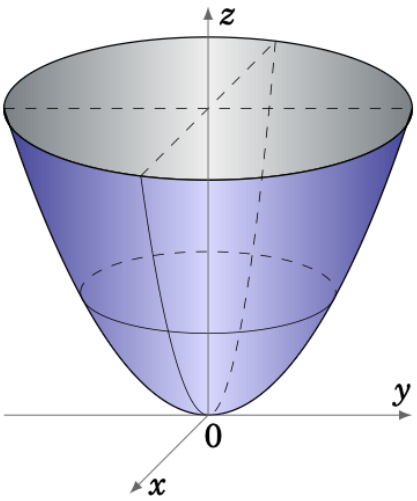


Figure 1.6.7 Paraboloid

An *elliptic cone* is the set

$$C = \left\{ (x, y, z) \in \mathbb{R}^3 : \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \right\},$$

where $a, b, c > 0$.

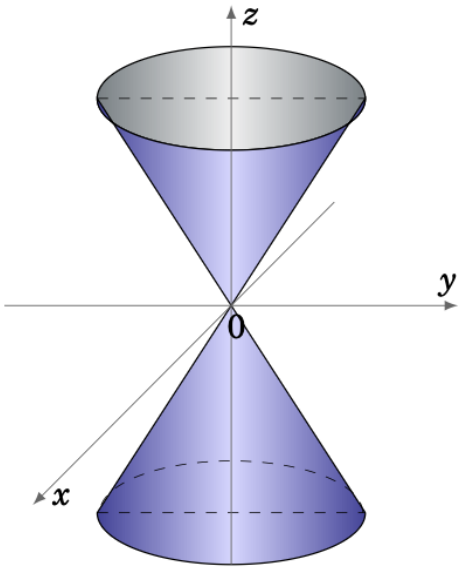


Figure 1.6.9 Elliptic cone

Lecture 2, October 9, 2026

1.4 Curvilinear coordinates

Polar coordinates. In Cartesian coordinates, a point in $\mathbb{R}^2$ is represented by an ordered pair (x, y), where x and y denote its coordinates along the x- and y-axes, respectively. An alternative way to describe the location of a point is by using its distance from the origin and the angle it makes with the positive x-axis. This leads to the notion of *polar coordinates*.

Let $P = (x, y)$ be a point in $\mathbb{R}^2$, different from the origin. Let r denote the distance from the origin to P, and let θ denote the angle measured counterclockwise from the positive x-axis to the line segment joining the origin to P (see Figure 9). Then the pair (r, θ) is called the *polar coordinates* of P.

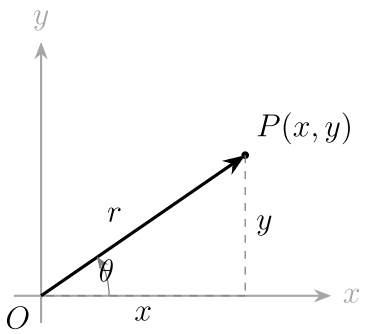


Figure 9: Polar coordinates in $\mathbb{R}^2$.

The Cartesian and polar coordinates are related by

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Conversely,

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right),$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we shall assume

$$r \geq 0, \qquad 0 \leq \theta < 2\pi.$$

Observe that the polar coordinates of the origin are not unique, since $r = 0$ while θ may be chosen arbitrarily.

Cartesian coordinates. We now return to three-dimensional Euclidean space. A point in $\mathbb{R}^3$ is represented by its Cartesian coordinates (x, y, z), which specify its position relative to the three coordinate axes. Geometrically, these coordinates may be obtained by starting at the origin, moving a distance x along the x-axis, then a distance y parallel to the y-axis, and finally a distance z parallel to the z-axis (see Figure 10).

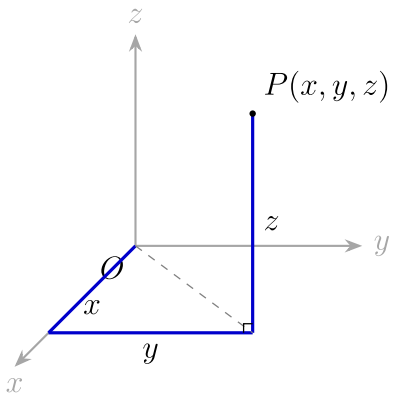


Figure 10: Cartesian coordinates in $\mathbb{R}^3$.

Although Cartesian coordinates are convenient for many problems, they are not always the most suitable choice. In many applications, the geometry of the problem possesses cylindrical or spherical symmetry. In such situations, it is often advantageous to describe a point using *curvilinear coordinates*, where one or more of the coordinates are determined by distances and angles rather than distances measured parallel to the coordinate axes.

Cylindrical coordinates. Let $P = (x, y, z)$ be a point in $\mathbb{R}^3$, and let $P_0 = (x, y, 0)$ denote the orthogonal projection of P onto the xy-plane. Treating (x, y) as a point in $\mathbb{R}^2$, let (r, θ) be its polar coordinates (see Figure 11). The triple (r, θ, z) is called the *cylindrical coordinates* of the point P.

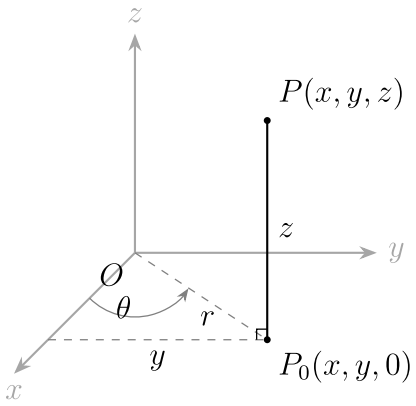


Figure 11: Cylindrical coordinates in $\mathbb{R}^3$.

The Cartesian and cylindrical coordinates are related by

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

Conversely,

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right), \qquad z = z,$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we assume

$$r \geq 0, \qquad 0 \leq \theta < 2\pi.$$

Observe that θ is not defined when $(x, y) = (0, 0)$, that is, for points lying on the z-axis.

Spherical coordinates. Let $P = (x, y, z)$ be a point in $\mathbb{R}^3$, different from the origin. Let ρ denote the distance from the origin to P, and let θ be the angle that the projection of $\overrightarrow{OP}$ onto the xy-plane makes with the positive x-axis. Finally, let ϕ be the angle between the positive z-axis and the vector $\overrightarrow{OP}$ (see Figure 12). The triple (ρ, θ, ϕ) is called the *spherical coordinates* of the point P.

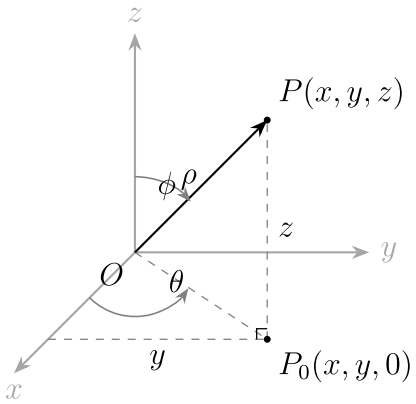


Figure 12: Spherical coordinates in $\mathbb{R}^3$.

The Cartesian and spherical coordinates are related by

$$x = \rho \sin\phi \cos\theta, \qquad y = \rho \sin\phi \sin\theta, \qquad z = \rho \cos\phi.$$

Conversely,

$$\rho = \sqrt{x^2 + y^2 + z^2}, \qquad \theta = \tan^{-1}\left(\frac{y}{x}\right), \qquad \phi = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right),$$

where the value of θ is chosen according to the quadrant in which the point (x, y) lies.

Throughout these notes, we assume

$$\rho \geq 0, \qquad 0 \leq \theta < 2\pi, \qquad 0 \leq \phi \leq \pi.$$

Observe that θ is not defined for points on the z-axis, while the spherical coordinates of the origin are not uniquely determined since $\rho = 0$.

2 Multivariable calculus

We now extend our study of functions to several variables. In particular, we will consider functions whose inputs and outputs may be vectors and develop the basic ideas needed to study them.

2.1 Vector-valued and scalar functions

Definition 7. *A vector-valued function of a real variable is a rule that associates a vector* $\mathbf{f}(t)$ *with a real number* t*, where* t *is in some subset* D *of* $\mathbb{R}^1$ *(called the* ***domain of*** $\mathbf{f}$*). We write* $\mathbf{f} : D \to \mathbb{R}^3$ *to denote that* $\mathbf{f}$ *is a mapping of* D *into* $\mathbb{R}^3$*.*

A vector-valued function of a real variable can be written in component form as

$$\boxed{\mathbf{f}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}}$$

or equivalently,

$$\boxed{\mathbf{f}(t) = \big(f_1(t), f_2(t), f_3(t)\big)},$$

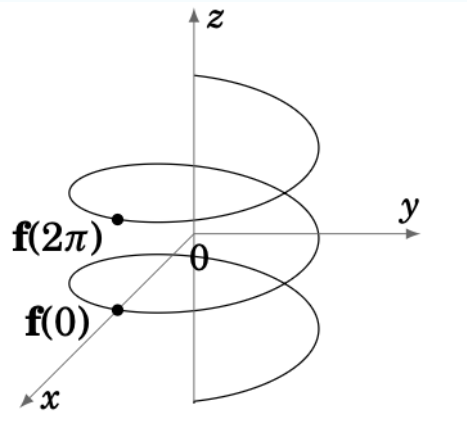
where $f_1(t)$, $f_2(t)$, and $f_3(t)$ are real-valued functions called the *component functions* of $\mathbf{f}$.

Example 1. *Define* $\mathbf{f} : \mathbb{R} \to \mathbb{R}^3$ *by*

$$\mathbf{f}(t) = (\cos t, \sin t, t).$$

This is the equation of a helix. *In particular,*

$$\mathbf{f}(0) = (1, 0, 0), \qquad \mathbf{f}(2\pi) = (1, 0, 2\pi).$$



Definition 8. *Let* $\mathbf{f}(t)$ *be a vector-valued function, let* a *be a real number, and let* $\mathbf{c}$ *be a vector. We say that*

$$\lim_{t\to a} \mathbf{f}(t) = \mathbf{c}$$

if

$$\lim_{t\to a} \|\mathbf{f}(t) - \mathbf{c}\| = 0.$$

If $\mathbf{f}(t) = (f_1(t), f_2(t), f_3(t))$, *then*

$$\lim_{t\to a} \mathbf{f}(t) = \left(\lim_{t\to a} f_1(t), \lim_{t\to a} f_2(t), \lim_{t\to a} f_3(t)\right),$$

provided that all three limits on the right-hand side exist.

Definition 9. *Let* $\mathbf{f}(t) = (f_1(t), f_2(t), f_3(t))$ *be a vector-valued function, and let* a *be a real number in its domain. Then* $\mathbf{f}(t)$ *is continuous at* a *if*

$$\lim_{t\to a} \mathbf{f}(t) = \mathbf{f}(a).$$

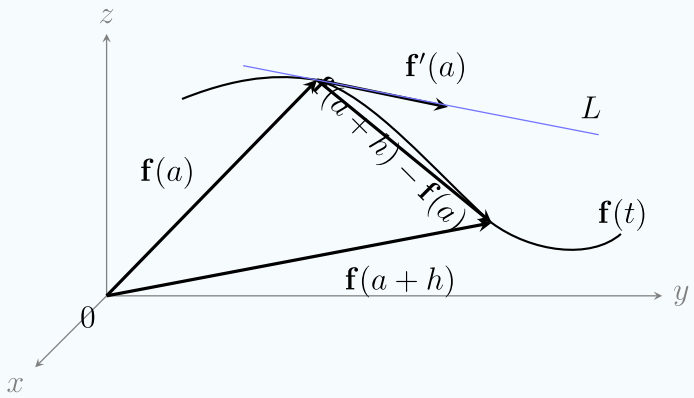
Equivalently, $\mathbf{f}(t)$ *is continuous at* a *if and only if* $f_1(t)$, $f_2(t)$, *and* $f_3(t)$ *are continuous at* a.
The derivative of $\mathbf{f}(t)$ *at* a, *denoted by* $\mathbf{f}'(a)$ *or* $\dfrac{d\mathbf{f}}{dt}(a)$, *is the limit*

$$\mathbf{f}'(a) = \lim_{h\to 0} \frac{\mathbf{f}(a+h) - \mathbf{f}(a)}{h},$$

if that limit exists. Equivalently,

$$\mathbf{f}'(a) = \big(f_1'(a), f_2'(a), f_3'(a)\big),$$

if the component derivatives exist. We say that $\mathbf{f}(t)$ *is differentiable at* a *if* $\mathbf{f}'(a)$ *exists.*



Recall that for a real-valued function of a single variable, the derivative represents the slope of the tangent line to the graph at a given point. For a vector-valued function, the derivative is a *tangent vector* to the curve represented by the function, and this vector lies along the *tangent line* to the curve at the corresponding point.

Example 2. *Let* $\mathbf{f}(t) = (\cos t, \sin t, t)$*. Then*

$$\mathbf{f}'(t) = (-\sin t, \cos t, 1)$$

for all t*. The tangent line* L *to the curve at* $\mathbf{f}(2\pi) = (1, 0, 2\pi)$ *is*

$$L = \mathbf{f}(2\pi) + s\mathbf{f}'(2\pi) = (1, 0, 2\pi) + s(0, 1, 1),$$

or, in parametric form,

$$x = 1, \qquad y = s, \qquad z = 2\pi + s, \qquad -\infty < s < \infty.$$

A *scalar function* is a real-valued function. If $u(t)$ is a scalar function and $\mathbf{f}(t)$ is a vector-valued function, then their product, defined by $(u\mathbf{f})(t) = u(t)\mathbf{f}(t)$, is a vector-valued function. The usual rules of differentiation and the basic properties of derivatives extend naturally to vector-valued functions.

Exercise. *Suppose* $\mathbf{f}(t)$ *is differentiable and* $\|\mathbf{f}(t)\| \neq 0$ *for all* t*. If* $\|\mathbf{f}(t)\|$ *is constant, show that*

$$\mathbf{f}(t) \cdot \mathbf{f}'(t) = 0 \qquad \text{for all } t.$$

2.2 Arc length

Arc length is useful in studying curvature and the motion of curves in differential geometry.

Let $\mathbf{r}(t) = (x(t), y(t), z(t))$ be the position vector of an object moving in $\mathbb{R}^3$. Its velocity is $\mathbf{r}'(t)$, so its speed at time t is $\|\mathbf{r}'(t)\|$. Therefore, the distance traveled by the object from $t = a$ to $t = b$ is

$$s = \int_a^b \|\mathbf{r}'(t)\|\, dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt.$$

This also gives the **arc length** of the curve traced by $\mathbf{r}(t)$ between $t = a$ and $t = b$.

Definition 10. *Let* C *be a smooth curve in* $\mathbb{R}^3$ *represented by a function* $\mathbf{f}(t)$ *defined on an interval* $[a, b]$*, and let* $\alpha : [c, d] \to [a, b]$ *be a smooth one-to-one mapping of an interval* $[c, d]$ *onto* $[a, b]$*. Then the function* $\mathbf{g} : [c, d] \to \mathbb{R}^3$ *defined by* $\mathbf{g}(s) = \mathbf{f}(\alpha(s))$ *is a* **parametrization** *of* C *with* parameter s*. If* α *is strictly increasing on* $[c, d]$*, then we say that* $\mathbf{g}(s)$ *is* **equivalent** *to* $\mathbf{f}(t)$*. A parametrization* $\mathbf{f} : [a, b] \to \mathbb{R}^3$ *is called* **smooth** *if* $\mathbf{f}'(t)$ *exists and* $\mathbf{f}'(t) \neq \mathbf{0}$ *for all* $t \in [a, b]$*.*

Theorem 5 (Chain Rule)**.** *If* $\mathbf{f}(t)$ *is a differentiable vector-valued function of* t*, and* $t = \alpha(s)$ *is a differentiable scalar function of* s*, then* $\mathbf{f}(s) = \mathbf{f}(\alpha(s))$ *is a differentiable vector-valued function of* s*, and*

$$\frac{d\mathbf{f}}{ds} = \frac{d\mathbf{f}}{dt}\frac{dt}{ds}$$

for any s *where the composite function* $\mathbf{f}(\alpha(s))$ *is defined.*

Example 3. *Let* $\mathbf{r}(t) = (\cos t, \sin t, t)$*,* $0 \leq t \leq 2\pi$*, and consider the reparametrization* $t = 2s$*. Then*

$$\mathbf{g}(s) = \mathbf{r}(2s) = (\cos 2s, \sin 2s, 2s), \qquad 0 \leq s \leq \pi.$$

By the chain rule,

$$\mathbf{g}'(s) = \mathbf{r}'(2s)\frac{dt}{ds} = 2(-\sin 2s, \cos 2s, 1).$$

Hence, the arc length of the curve using the parameter s is

$$L = \int_0^{\pi} \|\mathbf{g}'(s)\|\, ds = \int_0^{\pi} 2\sqrt{2}\, ds = 2\pi\sqrt{2}.$$

The same curve using the original parameter gives

$$L = \int_0^{2\pi} \|\mathbf{r}'(t)\|\, dt = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}.$$

Thus, equivalent parametrization gives the same arc length.

Example 4. *Consider the curve $\mathbf{r}(t) = (t, t^2, 0)$, $0 \le t \le 1$, and re-parametrize it by $t = s^3$, $0 \le s \le 1$. Then*

$$\mathbf{g}(s) = \mathbf{r}(s^3) = (s^3, s^6, 0).$$

Since $t = s^3$ is strictly increasing, $\mathbf{g}(s)$ is equivalent to $\mathbf{r}(t)$. By the chain rule,

$$\mathbf{g}'(s) = \mathbf{r}'(s^3)\frac{dt}{ds} = (1, 2s^3, 0)(3s^2) = (3s^2, 6s^5, 0).$$

Therefore, the arc length computed using the new parameter is

$$L = \int_0^1 \|\mathbf{g}'(s)\|\, ds = \int_0^1 3s^2\sqrt{1 + 4s^6}\, ds.$$

On the other hand, using the original parameter,

$$L = \int_0^1 \|\mathbf{r}'(t)\|\, dt = \int_0^1 \sqrt{1 + 4t^2}\, dt.$$

With the substitution $t = s^3$, these two expressions are equal, showing again that the arc length is independent of the choice of equivalent parametrization.

A curve may have different parametrizations, and the speed may depend on the choice of parameter. It is therefore useful to use the *arc length* as a parameter. Let $\mathbf{f}(t)$ be a smooth parametrization of a curve on $[a, b]$. Define

$$s = s(t) = \int_a^t \|\mathbf{f}'(u)\|\, du.$$

Here s represents the distance traveled along the curve from $t = a$ to t. In particular,

$$s(a) = 0, \qquad s(b) = L,$$

where L is the total arc length of the curve. By the Fundamental Theorem of Calculus,

$$s'(t) = \|\mathbf{f}'(t)\| > 0.$$

Hence $s(t)$ is strictly increasing and has a differentiable inverse $t = \alpha(s)$, where

$$\alpha'(s) = \frac{1}{\|\mathbf{f}'(\alpha(s))\|}.$$

We can therefore reparametrize the curve using the arc length parameter s by defining

$$\mathbf{f}(s) = \mathbf{f}(\alpha(s)), \qquad 0 \le s \le L.$$

By the Chain Rule,

$$\mathbf{f}'(s) = \mathbf{f}'(\alpha(s))\alpha'(s) = \frac{\mathbf{f}'(\alpha(s))}{\|\mathbf{f}'(\alpha(s))\|}.$$

Consequently,

$$\|\mathbf{f}'(s)\| = 1.$$

Thus, an **arc length parametrization** traverses the curve with ***unit speed***.

Theorem 6 (Inverse Function Theorem)**.** *Let $s : [a, b] \to \mathbb{R}$ be differentiable and one-to-one, with $s'(t) \neq 0$ for all $t \in [a, b]$. Then s has an inverse function*

$$t = \alpha(s),$$

defined on $s([a, b])$. The inverse function α is differentiable, and

$$\alpha'(s) = \frac{1}{s'(\alpha(s))}.$$

Example 5. *Parametrize the curve*

$$\mathbf{f}(t) = (\cos t, \sin t, 2t), \qquad t \in [0, 2\pi],$$

by arc length.
Solution: *We have*

$$\mathbf{f}'(t) = (-\sin t, \cos t, 2),$$

and hence

$$\|\mathbf{f}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 4} = \sqrt{5}.$$

Therefore, by the definition of arc length parameter,

$$s = \int_0^t \|\mathbf{f}'(u)\|\, du = \int_0^t \sqrt{5}\, du = \sqrt{5}\, t$$

for all $t \in [0, 2\pi]$.

Thus,

$$t = \alpha(s) = \frac{s}{\sqrt{5}}.$$

Since $t \in [0, 2\pi]$, the corresponding range of s is

$$0 \le s \le 2\pi\sqrt{5}.$$

Hence the required arc length parametrization is

$$\mathbf{f}(s) = \left(\cos\frac{s}{\sqrt{5}}, \sin\frac{s}{\sqrt{5}}, \frac{2s}{\sqrt{5}}\right), \qquad s \in [0, 2\pi\sqrt{5}].$$

Indeed,

$$\|\mathbf{f}'(s)\| = 1.$$

For a smooth curve $\mathbf{f}(t)$ with $\mathbf{f}'(t) \neq \mathbf{0}$, the vector $\mathbf{f}'(t)$ gives the **tangent direction** to the curve. The corresponding **unit tangent vector** is

$$\mathbf{T}(t) = \frac{\mathbf{f}'(t)}{\|\mathbf{f}'(t)\|}.$$

The direction in which the tangent vector changes is described by the **principal normal vector**. Its unit version is given by

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \frac{\mathbf{f}'(t) \times \big(\mathbf{f}''(t) \times \mathbf{f}'(t)\big)}{\|\mathbf{f}'(t)\| \, \|\mathbf{f}''(t) \times \mathbf{f}'(t)\|}.$$

The **unit binormal vector** is defined by

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \frac{\mathbf{f}'(t) \times \mathbf{f}''(t)}{\|\mathbf{f}'(t) \times \mathbf{f}''(t)\|}.$$

Thus, $\mathbf{T}(t)$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$ form a right-handed system of mutually perpendicular unit vectors, called an **orthonormal frame**, attached to the curve at each point. Finally, the **curvature** $\kappa(t)$ measures how much the curve bends at a given point and is given by

$$\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{f}'(t)\|} = \frac{\|\mathbf{f}'(t) \times \mathbf{f}''(t)\|}{\|\mathbf{f}'(t)\|^3}.$$

Thus, a larger value of $\kappa(t)$ indicates a sharper bend, while a smaller value indicates that the curve is flatter. Check these quantities explicitly for the helix $\mathbf{f}(t) = (\cos t, \sin t, t)$!

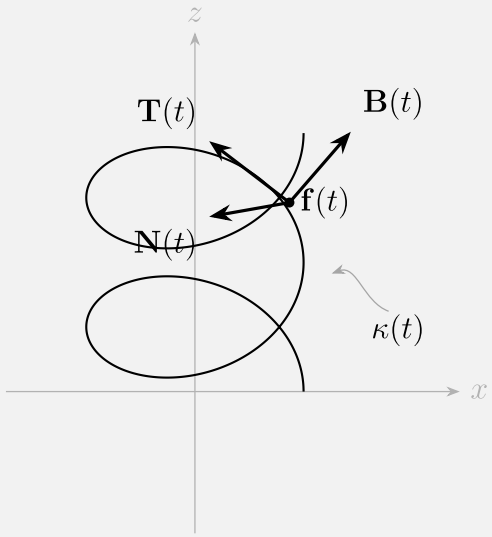


Figure 13: Frenet frame along a curve

Lecture 3, October 12, 2026

2.3 Functions of two or three variables

A real-valued function f defined on a subset D of $\mathbb{R}^2$ is a rule that assigns to each point (x, y) in D a real number $f(x, y)$. The largest possible set D in $\mathbb{R}^2$ on which f is defined is called the **domain** of f, and the **range** of f is the set of all real numbers $f(x, y)$ as (x, y) varies over the domain D. A similar definition holds for functions $f(x, y, z)$ defined on points (x, y, z) in $\mathbb{R}^3$.

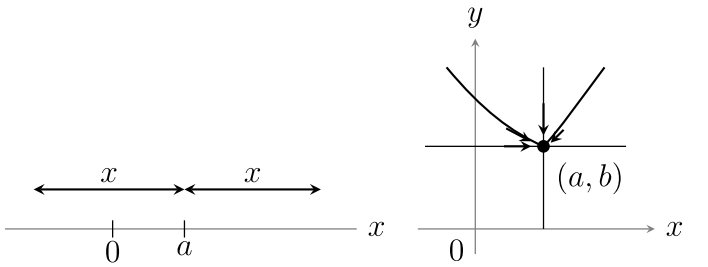
Example 6. *Consider the following functions and determine their domains and ranges.*

1. *Let* $f(x, y) = \dfrac{1}{x - y}$. *The domain is* $D = \{(x, y) \in \mathbb{R}^2 : x \neq y\}$, *and the range is* $\mathbb{R} \setminus \{0\}$.
2. *Let* $f(x, y) = \sqrt{1 - x^2 - y^2}$. *The domain is* $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$, *and the range is* $[0, 1]$.
3. *Let* $f(x, y, z) = e^{x+y-z}$. *The domain is* $D = \mathbb{R}^3$, *and the range is* $(0, \infty)$.

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function of two variables. We usually write $z = f(x, y)$. The **graph** of f is the set of points $G_f = \{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$. Thus, the graph of a function of two variables is a surface in $\mathbb{R}^3$. Equivalently, if we define $F(x, y, z) =$

$f(x, y) - z$, then the graph can be written as $G_f = \{(x, y, z) \in \mathbb{R}^3 : F(x, y, z) = 0\}$.

In one variable, when we write $x \to a$, the point x can approach a only from the two directions along the real line, as illustrated in (a) below. In two variables, however, a point (x, y) can approach (a, b) from infinitely many directions and along infinitely many paths, as shown in (b) below. Therefore, the idea of a limit in several variables must account for all possible ways of approaching the point. To capture this geometrically, instead of considering an interval around a, we consider a disk centered at (a, b). If (x, y) is sufficiently close to (a, b), it lies inside this disk regardless of the direction or path along which it approaches (a, b). This motivates the definition of the limit of a function of several variables: the value of the function should approach the same number as (x, y) approaches (a, b) from every possible direction and along every possible path.



(a) $x \to a$ in $\mathbb{R}$ **(b)** $(x, y) \to (a, b)$ in $\mathbb{R}^2$

Definition 11. *Let $(a, b) \in \mathbb{R}^2$ and let $\delta > 0$. The **deleted neighborhood** of (a, b) with radius δ is the set of all points $(x, y) \in \mathbb{R}^2$ satisfying*

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta.$$

Thus, it consists of all points inside the disk of radius δ centered at (a, b), except the point (a, b) itself.

Definition 12. *Let $(a, b) \in \mathbb{R}^2$, and let f be a real-valued function defined at all points (x, y) in some deleted neighborhood of (a, b). We say that the limit of $f(x, y)$ as (x, y) approaches (a, b) is L, written as*

$$\lim_{(x,y)\to(a,b)} f(x, y) = L,$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(x, y) - L| < \epsilon$$

whenever

$$0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta.$$

In many simple cases, the limit of a function of several variables can be found by direct substitution. If $f(x, y)$ is defined at (a, b) and its formula does not lead to an indeterminate form $0/0$ at that point, then

$$\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b).$$

Example 7. *Consider*

$$\lim_{(x,y)\to(1,2)} \frac{xy}{x^2 + y^2}.$$

Since the function is defined at $(1, 2)$*, we can substitute* $x = 1$ *and* $y = 2$ *directly:*

$$\lim_{(x,y)\to(1,2)} \frac{xy}{x^2 + y^2} = \frac{(1)(2)}{1^2 + 2^2} = \frac{2}{5}.$$

Example 8. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2 + y^2}.$$

Direct substitution gives the indeterminate form $0/0$*. To check whether the limit exists, consider different paths approaching* $(0, 0)$*.*
Along the x*-axis,* $y = 0$*, so*

$$\frac{xy}{x^2 + y^2} = 0.$$

Along the line $y = x$*, we obtain*

$$\frac{xy}{x^2 + y^2} = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Since the function approaches different values along these two paths, the limit does not exist.

Theorem 7. *Suppose that* $\lim_{(x,y)\to(a,b)} f(x, y)$ *and* $\lim_{(x,y)\to(a,b)} g(x, y)$ *both exist, and that* k *is some scalar. Then:*

(a)

$$\lim_{(x,y)\to(a,b)} [f(x, y) \pm g(x, y)] = \left[\lim_{(x,y)\to(a,b)} f(x, y)\right] \pm \left[\lim_{(x,y)\to(a,b)} g(x, y)\right].$$

(b)

$$\lim_{(x,y)\to(a,b)} kf(x, y) = k\left[\lim_{(x,y)\to(a,b)} f(x, y)\right].$$

(c)

$$\lim_{(x,y)\to(a,b)}[f(x,y)g(x,y)] = \left[\lim_{(x,y)\to(a,b)} f(x,y)\right]\left[\lim_{(x,y)\to(a,b)} g(x,y)\right].$$

(d)

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y)}{g(x,y)} = \frac{\lim_{(x,y)\to(a,b)} f(x,y)}{\lim_{(x,y)\to(a,b)} g(x,y)} \qquad \text{if} \qquad \lim_{(x,y)\to(a,b)} g(x,y) \neq 0.$$

(e) If $|f(x,y) - L| \leq g(x,y)$ *for all* (x,y) *and if*

$$\lim_{(x,y)\to(a,b)} g(x,y) = 0,$$

then

$$\lim_{(x,y)\to(a,b)} f(x,y) = L.$$

Example 9. *Show that*

$$\lim_{(x,y)\to(0,0)} \frac{x^4}{x^2+y^2} = 0.$$

Since direct substitution gives the indeterminate form $0/0$*, we use Theorem 7(e). For all* $(x,y) \neq (0,0)$*,*

$$0 \leq \left|\frac{x^4}{x^2+y^2}\right| \leq \frac{(x^2+y^2)^2}{x^2+y^2} = x^2+y^2 \to 0$$

as $(x,y) \to (0,0)$*, where we used* $x^4 \leq (\sqrt{x^2})^4 \leq (\sqrt{x^2+y^2})^4$ *for all* (x,y)*. Therefore,*

$$\lim_{(x,y)\to(0,0)} \frac{x^4}{x^2+y^2} = 0.$$

Definition 13. *A real-valued function* $f(x,y)$ *with domain* D *in* $\mathbb{R}^2$ *is* ***continuous*** *at the point* (a,b) *in* D *if*

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

We say that $f(x,y)$ *is a* ***continuous function*** *if it is continuous at every point in its domain* D*.*

Example 10. *Define* $f : \mathbb{R}^2 \to \mathbb{R}$ *by*

$$f(x,y) = \begin{cases} \dfrac{x^2 y}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

For $(x,y) \neq (0,0)$*, we have*

$$\left| \frac{x^2 y}{x^2+y^2} \right| \leq |y| \leq \sqrt{x^2+y^2} \to 0$$

as $(x,y) \to (0,0)$*. Hence*

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0 = f(0,0).$$

Therefore, f *is continuous at* $(0,0)$*. At every other point,* f *is continuous by direct substitution. Thus,* f *is continuous on* $\mathbb{R}^2$*.*

Example 11. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{x}{y}.$$

Direct substitution gives the indeterminate form $0/0$*. To check the limit, consider different paths approaching* $(0,0)$*. Along the line* $y = x$*,*

$$\frac{x}{y} = 1,$$

whereas along the line $y = 2x$*,*

$$\frac{x}{y} = \frac{1}{2}.$$

Since the function approaches different values along different paths, the limit does not exist.

Example 12. *Consider*

$$\lim_{(x,y)\to(0,0)} \frac{y^4 \sin(xy)}{x^2+y^2}.$$

Since $|\sin(xy)| \leq 1$*,*

$$\left| \frac{y^4 \sin(xy)}{x^2+y^2} \right| \leq \frac{y^4}{x^2+y^2} \leq \frac{y^2(y^2+x^2)}{x^2+y^2} \leq y^2 \to 0$$

as $(x,y) \to (0,0)$*. Therefore,*

$$\lim_{(x,y)\to(0,0)} \frac{y^4 \sin(xy)}{x^2+y^2} = 0.$$

Example 13. *Use the substitution* $r = \sqrt{x^2 + y^2}$ *to show that*

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} = 1.$$

Solution: *Let* $r = \sqrt{x^2 + y^2}$. *Then* $(x, y) \to (0, 0)$ *implies* $r \to 0$, *and the limit becomes*

$$\lim_{r\to 0} \frac{\sin r}{r} = 1.$$

This is the standard single-variable limit, which can be obtained, for example, using L'Hôpital's Rule:

$$\lim_{r\to 0} \frac{\sin r}{r} = \lim_{r\to 0} \frac{\cos r}{1} = 1.$$

Hence,

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} = 1.$$

2.4 Partial derivatives

When a function depends on several variables, we may want to study how it changes with respect to one variable while keeping the other variables fixed. This leads to the idea of a **partial derivative**. For a function $f(x, y)$, we can therefore differentiate with respect to x while keeping y fixed, or with respect to y while keeping x fixed.

Definition 14. *Let* $f(x, y)$ *be a real-valued function with domain* D *in* $\mathbb{R}^2$, *and let* (a, b) *be a point in* D. *Then the* ***partial derivative*** *of* f *at* (a, b) *with respect to* x, *denoted by* $\dfrac{\partial f}{\partial x}(a, b)$, *is defined as*

$$\frac{\partial f}{\partial x}(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}.$$

The ***partial derivative*** *of* f *at* (a, b) *with respect to* y, *denoted by* $\dfrac{\partial f}{\partial y}(a, b)$, *is defined as*

$$\frac{\partial f}{\partial y}(a,b) = \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}.$$

Since $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ are themselves functions of x and y, we can differentiate them again.

This gives the **higher-order partial derivatives**. For example,

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = f_{xx} = D_{11}f, \qquad \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = f_{yy} = D_{22}f,$$

and the mixed partial derivatives are

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = f_{xy} = D_{12}f, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = f_{yx} = D_{21}f.$$

Higher-order partial derivatives are obtained similarly by repeatedly differentiating.

Example 14. *Let*

$$f(x, y) = x^2 - y^2 + 6xy + 4x - 8y + 2.$$

The first-order partial derivatives are

$$\frac{\partial f}{\partial x} = 2x + 6y + 4, \qquad \frac{\partial f}{\partial y} = -2y + 6x - 8.$$

The second-order partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = 2, \qquad \frac{\partial^2 f}{\partial y^2} = -2,$$

and

$$\frac{\partial^2 f}{\partial y\,\partial x} = 6, \qquad \frac{\partial^2 f}{\partial x\,\partial y} = 6.$$

Since the second-order mixed partial derivatives are equal, we have

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y} = 6.$$

The third-order partial derivatives are all zero:

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial^3 f}{\partial y^3} = \frac{\partial^3 f}{\partial y\,\partial x^2} = \frac{\partial^3 f}{\partial x\,\partial y^2} = \frac{\partial^3 f}{\partial y^2\,\partial x} = \frac{\partial^3 f}{\partial x^2\,\partial y} = \frac{\partial^3 f}{\partial x\,\partial y\,\partial x} = \frac{\partial^3 f}{\partial y\,\partial x\,\partial y} = 0.$$

In general, the mixed partial derivatives f_{xy} and f_{yx} need not be equal. However, if the second-order partial derivatives f_{xy} and f_{yx} are continuous in a neighborhood of a point, then

$$f_{xy} = f_{yx}$$

at that point. This result is known as **Clairaut's theorem** (or **Schwarz's theorem**). The equality need not hold when the required continuity condition is absent. For

example, define

$$f(x,y) = \begin{cases} \dfrac{xy(x^2-y^2)}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Then, at $(0,0)$,

$$f_{xy}(0,0) = -1, \qquad f_{yx}(0,0) = 1.$$

Thus,

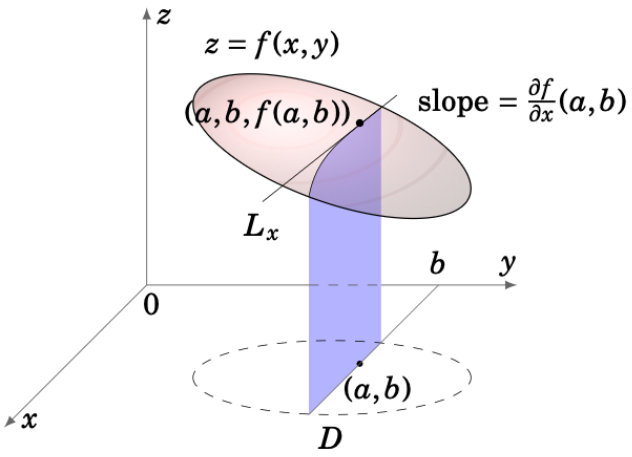
$$f_{xy}(0,0) \neq f_{yx}(0,0).$$

2.5 Tangent plane to a surface

Recall that the derivative $\dfrac{dy}{dx}$ of a function $y = f(x)$ has a geometric meaning, namely, as the slope of the tangent line to the graph of f at the point $(x, f(x))$ in $\mathbb{R}^2$. In particular, at $x = a$, the slope of the tangent line is $f'(a)$. Since the tangent line passes through $(a, f(a))$, its equation is

$$y - f(a) = f'(a)(x - a).$$

We now seek a similar geometric interpretation for the partial derivatives of a function of two variables. The partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ can be thought of as the rate of change of a function $z = f(x,y)$ in the positive x and y directions, respectively. Let $(a, b, f(a,b))$ be a point on the surface $z = f(x,y)$. Then we want to find the equation of the tangent plane at that point. First, look at the image (a).



(a) Tangent line L_x in the plane $y = b$

The equation of the tangent line L_x will be as follows: This line stays in the plane $y = b$,

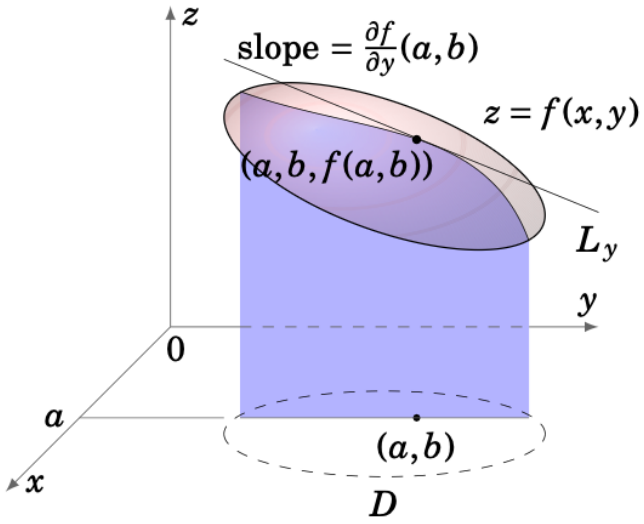
and so x and z vary. Therefore, the equation will be

$$z - f(a,b) = \frac{\partial f}{\partial x}(a,b)(x-a), \qquad y = b.$$

Hence, in the vector equation form, it can be written as

$$(x,y,z) = (a,b,f(a,b)) + t(1,0,\frac{\partial f}{\partial x}(a,b)).$$

Now, look at the second image (b).

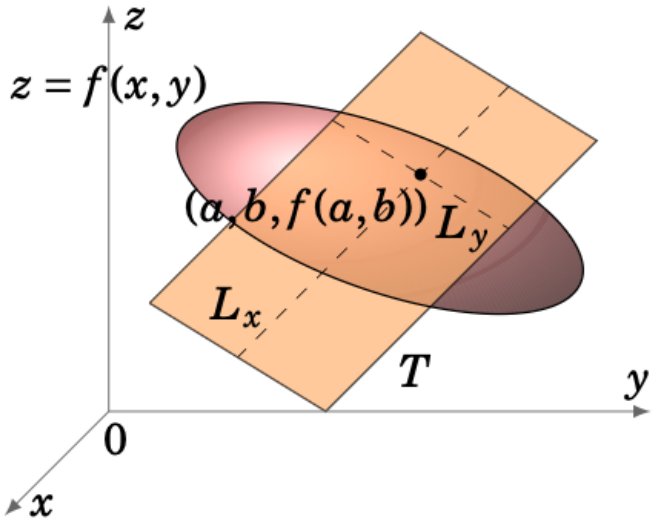


(b) Tangent line L_y in the plane $x = a$

So, in the vector equation form, the equation of L_y can be written as

$$(x,y,z) = (a,b,f(a,b)) + t(0,1,\frac{\partial f}{\partial y}(a,b)).$$

Now the tangent plane is



This contains the lines L_x and L_y. Then, it contains the vectors

$$v_x = (1, 0, \frac{\partial f}{\partial x}(a, b)) \qquad \text{and} \qquad v_y = (0, 1, \frac{\partial f}{\partial y}(a, b)),$$

and so it is perpendicular to the normal vector

$$\mathbf{n} = v_x \times v_y = \left(-\frac{\partial f}{\partial x}(a, b), -\frac{\partial f}{\partial y}(a, b), 1 \right).$$

Hence, in terms of the vector equation, the tangent plane will be

$$\mathbf{n} \cdot ((x, y, z) - (a, b, f(a, b))) = 0.$$

So, the equation of the tangent plane to the surface $z = f(x, y)$ at the point $(a, b, f(a, b))$ is

$$\frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b) - z + f(a, b) = 0.$$

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Suppose a surface is defined implicitly by

$$F(x, y, z) = 0$$

and passes through (a, b, c). The gradient

$$\nabla F(a, b, c) = \big(F_x(a, b, c), F_y(a, b, c), F_z(a, b, c)\big)$$

is **normal to the surface** at (a, b, c). Hence, the tangent plane is

$$\boxed{\nabla F(a, b, c) \cdot (x - a, y - b, z - c) = 0,}$$

or equivalently,

$$\boxed{F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0.}$$

Example 15. *Find the equation of the tangent plane to the surface $z = xe^y$ at the point $(1, 0, 1)$.*

Solution: *For $f(x, y) = xe^y$, we have $\frac{\partial f}{\partial x} = e^y$ and $\frac{\partial f}{\partial y} = xe^y$. Thus, at $(1, 0)$,*

$f_x(1,0) = 1$ *and* $f_y(1,0) = 1$. *Therefore, the tangent plane is*

$$(x-1) + (y-0) - z + 1 = 0,$$

or

$$x + y - z = 0.$$

Example 16. *Find the equation of the tangent plane to the surface* $x^2 + y^2 - z^2 = 0$ *at the point* $P = (3,4,5)$.
Solution: *Let* $F(x,y,z) = x^2 + y^2 - z^2$. *Then* $F_x = 2x$, $F_y = 2y$, *and* $F_z = -2z$. *At* $P = (3,4,5)$, *we have* $F_x = 6$, $F_y = 8$, *and* $F_z = -10$. *Therefore, the tangent plane is*

$$6(x-3) + 8(y-4) - 10(z-5) = 0,$$

or

$$3x + 4y - 5z = 0.$$

2.6 Directional derivatives and the gradient

For a function $z = f(x,y)$, the partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ describe the instantaneous rate of change of f in the positive x- and y-directions, respectively. But what if we want to measure the rate of change in an arbitrary direction? This leads to the idea of the **directional derivative**, which gives the rate of change of f in any specified direction. Thus, while partial derivatives describe changes along the coordinate directions, directional derivatives allow us to study changes along any direction in the plane.

Definition 15. *Let* $f(x,y)$ *be a real-valued function with domain* D *in* $\mathbb{R}^2$, *and let* (a,b) *be a point in* D. *Let* $\mathbf{v}$ *be a unit vector in* $\mathbb{R}^2$. *The* ***directional derivative*** *of* f *at* (a,b) *in the direction of* $\mathbf{v}$, *denoted by* $D_{\mathbf{v}}f(a,b)$, *is defined as*

$$D_{\mathbf{v}}f(a,b) = \lim_{h \to 0} \frac{f((a,b) + h\mathbf{v}) - f(a,b)}{h}.$$

If $f(x,y)$ has continuous partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ in D, then the directional derivative has a simple formula. Let us first recall the **Mean Value Theorem**: If g is continuous on $[a,b]$ and differentiable on (a,b), then there exists $\theta \in (0,1)$ such that

$$g(b) - g(a) = g'\big(a + \theta(b-a)\big)(b-a).$$

Theorem 8. *Let $f(x, y)$ be a real-valued function with domain D in $\mathbb{R}^2$ such that the partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist and are continuous in D. Let (a, b) be a point in D, and let $\mathbf{v} = (v_1, v_2)$ be a unit vector in $\mathbb{R}^2$. Then*

$$D_{\mathbf{v}} f(a, b) = v_1 \frac{\partial f}{\partial x}(a, b) + v_2 \frac{\partial f}{\partial y}(a, b).$$

Proof. Let $\mathbf{v} = (v_1, v_2)$ and fix $h \neq 0$. We split the change in f into two parts:

$$f(a+hv_1, b+hv_2) - f(a, b) = \big[f(a+hv_1, b+hv_2) - f(a+hv_1, b)\big] + \big[f(a+hv_1, b) - f(a, b)\big].$$

For the first part, keep $x = a + hv_1$ fixed and apply the Mean Value Theorem to the function of one variable obtained by varying y. Thus, for some number α between 0 and 1,

$$f(a + hv_1, b + hv_2) - f(a + hv_1, b) = hv_2 \frac{\partial f}{\partial y}(a + hv_1, b + \alpha hv_2).$$

Similarly, applying the Mean Value Theorem to the second part, there exists a number β between 0 and 1 such that

$$f(a + hv_1, b) - f(a, b) = hv_1 \frac{\partial f}{\partial x}(a + \beta hv_1, b).$$

Therefore,

$$\frac{f(a + hv_1, b + hv_2) - f(a, b)}{h} = v_2 \frac{\partial f}{\partial y}(a + hv_1, b + \alpha hv_2) + v_1 \frac{\partial f}{\partial x}(a + \beta hv_1, b).$$

As $h \to 0$, both points on the right approach (a, b). Since the partial derivatives are continuous,

$$D_{\mathbf{v}} f(a, b) = v_1 \frac{\partial f}{\partial x}(a, b) + v_2 \frac{\partial f}{\partial y}(a, b).$$

□

The directional derivative can be written in a particularly simple form using the **gradient**. For a real-valued function $f(x, y)$, the gradient is the vector

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

Similarly, for a function $f(x, y, z)$, the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right).$$

The symbol ∇ is called the **del operator**. Since the gradient contains all the first-order partial derivatives, the directional derivative in the direction of a unit vector $\mathbf{v}$

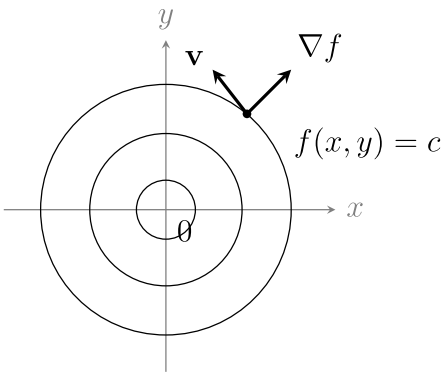
can be written as

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f.$$

Example 17. *Find the directional derivative of* $f(x, y) = x^2 + xy + y^2$ *at the point* $(1, -1)$ *in the direction of* $\mathbf{v} = (2, 1)$.
Solution: *Always normalize first. The unit vector in the given direction is* $\mathbf{u} = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$. *Since* $\nabla f = (2x + y, x + 2y)$, *we have* $\nabla f(1, -1) = (1, -1)$. *Therefore,*

$$D_{\mathbf{u}} f(1, -1) = \mathbf{u} \cdot \nabla f(1, -1) = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) \cdot (1, -1) = \frac{1}{\sqrt{5}}.$$

A real-valued function $z = f(x, y)$ whose partial derivatives $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist and are continuous is called **continuously differentiable**. Assume that $f(x, y)$ is continuously differentiable and that $\nabla f \neq \mathbf{0}$. Let c be a real number in the range of f, and let $\mathbf{v}$ be a unit vector tangent to the level curve $f(x, y) = c$, as shown in the figure below.



Since $f(x, y)$ is constant along a level curve $f(x, y) = c$, its rate of change in any tangent direction to the curve is zero. But

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f = \|\mathbf{v}\| \, \|\nabla f\| \cos\theta,$$

where θ is the angle between $\mathbf{v}$ and ∇f. For a unit tangent vector $\mathbf{v}$, we have $\|\mathbf{v}\| = 1$, and since $\nabla f \neq \mathbf{0}$, the condition $D_{\mathbf{v}} f = 0$ gives $\cos\theta = 0$. Hence $\theta = 90°$, so ∇f is perpendicular to the level curve. Thus, the gradient gives the normal direction to a level curve.

Moreover, for any unit vector $\mathbf{v}$, we have $D_{\mathbf{v}} f = \|\nabla f\| \cos\theta$. At a fixed point, $\|\nabla f\|$ is fixed, so the directional derivative is largest when $\theta = 0°$, that is, in the direction of ∇f, and smallest when $\theta = 180°$, that is, in the direction of $-\nabla f$. Therefore, f increases most rapidly in the direction of ∇f and decreases most rapidly in the direction of $-\nabla f$.

Although the above discussion is for functions of two variables, the same ideas extend naturally to functions of three or more variables. In particular, for a function $f(x, y, z)$, the

directional derivative in the direction of a unit vector $\mathbf{v}$ is given by

$$D_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f.$$

The same interpretation of the gradient therefore remains valid: ∇f gives the direction of fastest increase, while $-\nabla f$ gives the direction of fastest decrease.

Theorem 9. *Let $f(x, y)$ be a continuously differentiable real-valued function, with $\nabla f \neq \mathbf{0}$. Then:*

(a) The gradient ∇f is normal to any level curve $f(x, y) = c$.

(b) The value of $f(x, y)$ increases fastest in the direction of ∇f.

(c) The value of $f(x, y)$ decreases fastest in the direction of $-\nabla f$.

Example 18. *The temperature at a point (x, y, z) in a room is given by $T(x, y, z) = x^2 + 2y^2 + z^2$. In which direction from the point $(1, -1, 2)$ will the temperature decrease most rapidly?*
Solution: *We have $\nabla T = (2x, 4y, 2z)$, so $\nabla T(1, -1, 2) = (2, -4, 4)$. Since the temperature decreases most rapidly in the direction of $-\nabla T$, the required direction is*

$$-\nabla T(1, -1, 2) = (-2, 4, -4).$$

The corresponding unit direction vector is

$$\frac{1}{6}(-2, 4, -4) = \left(-\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right).$$

2.7 Maxima and minima

The gradient can also be used to find **extreme points** of a real-valued function, that is, points where the function has a **local maximum** or **local minimum**. We will focus on functions of two variables. The functions of three or more variables require methods using linear algebra, such as matrices, quadratic forms, and eigenvalues and eigenvectors.

Since the gradient describes the direction of fastest increase of a function, it is also useful for locating points where the function does not increase or decrease in any direction. This leads naturally to the study of local maxima and minima.

Definition 16. *Let $f(x, y)$ be a real-valued function, and let (a, b) be a point in the domain of f. We say that f has a **local maximum** at (a, b) if there exists $r > 0$*

such that $f(x,y) \le f(a,b)$ *whenever* $(x-a)^2+(y-b)^2 < r^2$. *Similarly,* f *has a* ***local minimum*** *at* (a,b) *if there exists* $r > 0$ *such that* $f(x,y) \ge f(a,b)$ *whenever* $(x-a)^2+(y-b)^2 < r^2$.
If $f(x,y) \le f(a,b)$ *for every* (x,y) *in the domain of* f, *then* f *has a* ***global maximum*** *at* (a,b). *Similarly, if* $f(x,y) \ge f(a,b)$ *for every* (x,y) *in the domain of* f, *then* f *has a* ***global minimum*** *at* (a,b).

Suppose that (a,b) is a local maximum or minimum point of $f(x,y)$. By fixing $y = b$, we obtain a single-variable function $g(x) = f(x,b)$, which has a local maximum or minimum at $x = a$. Hence, $g'(a) = 0$, giving $\dfrac{\partial f}{\partial x}(a,b) = 0$. Similarly, by fixing $x = a$, we obtain $\dfrac{\partial f}{\partial y}(a,b) = 0$. Therefore,

$$\nabla f(a,b) = \mathbf{0}.$$

Theorem 10. *Let* $f(x,y)$ *be a real-valued function such that* $\dfrac{\partial f}{\partial x}(a,b)$ *and* $\dfrac{\partial f}{\partial y}(a,b)$ *exist. A necessary condition for* f *to have a local maximum or minimum at* (a,b) *is*

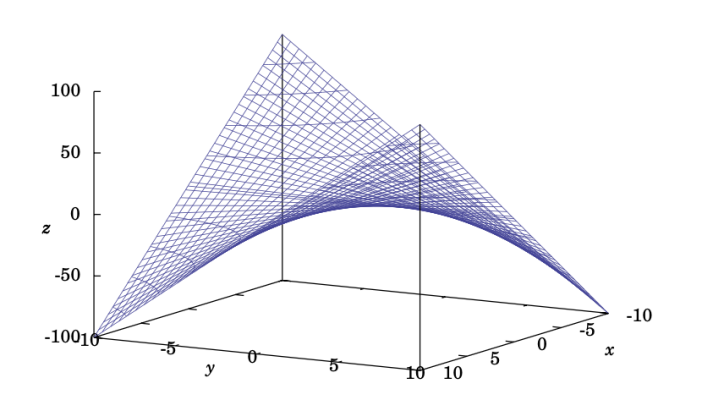
$$\nabla f(a,b) = \mathbf{0}.$$

A point (a,b) satisfying $\nabla f(a,b) = \mathbf{0}$ is called a **critical point** of f. Thus, to find the critical points, we solve

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0$$

simultaneously. Note that $\nabla f(a,b) = \mathbf{0}$ is only a **necessary condition**; a critical point need not be a local maximum or minimum.

Example 19. *Consider the function* $f(x,y) = xy$. *Then* $\dfrac{\partial f}{\partial x} = y$ *and* $\dfrac{\partial f}{\partial y} = x$, *so the only critical point is* $(0,0)$. *However,* $(0,0)$ *is neither a local maximum nor a local minimum. Along the path* $y = x$, *we have* $f(x,y) = x^2 > 0 = f(0,0)$ *for* $x \neq 0$, *while along the path* $y = -x$, *we have* $f(x,y) = -x^2 < 0 = f(0,0)$. *Thus, the function takes both larger and smaller values arbitrarily close to* $(0,0)$.
Hence, $(0,0)$ *is a* ***saddle point****: the function increases in some directions and decreases in other directions. The graph of* $f(x,y)$ *is shown in the figure.*



Theorem 11. *Let $f(x,y)$ be a twice continuously differentiable real-valued function, with a critical point at (a,b), that is, $\nabla f(a,b) = \mathbf{0}$. Define*

$$D = \frac{\partial^2 f}{\partial x^2}(a,b)\frac{\partial^2 f}{\partial y^2}(a,b) - \left(\frac{\partial^2 f}{\partial y \partial x}(a,b)\right)^2.$$

Then

(a) if $D > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(a,b) > 0$, then f has a local minimum at (a,b);

(b) if $D > 0$ and $\dfrac{\partial^2 f}{\partial x^2}(a,b) < 0$, then f has a local maximum at (a,b);

*(c) if $D < 0$, then f has a **saddle point** at (a,b);*

(d) if $D = 0$, then the test fails.

Example 20. *Find all local maxima and minima of*

$$f(x,y) = x^3 + y^3 - 3xy.$$

Solution: *We first find the critical points. Since $f_x = 3x^2 - 3y$ and $f_y = 3y^2 - 3x$, the critical points satisfy*

$$x^2 - y = 0, \qquad y^2 - x = 0.$$

Thus $y = x^2$ and $x = y^2$, giving the critical points $(0,0)$ and $(1,1)$.
The second-order partial derivatives are $f_{xx} = 6x$, $f_{yy} = 6y$, and $f_{yx} = -3$. Hence

$$D = f_{xx}f_{yy} - (f_{yx})^2 = 36xy - 9.$$

*At $(0,0)$, $D = -9 < 0$, so $(0,0)$ is a **saddle point**.*

At $(1,1)$*,* $D = 27 > 0$ *and* $f_{xx}(1,1) = 6 > 0$*. Therefore,* $(1,1)$ *is a* ***local minimum****. Hence,* f *has a local minimum at* $(1,1)$ *and a saddle point at* $(0,0)$*, and it has no local maximum.*

Lecture 5, October 16, 2026

3 Multiple integrals

Multiple integrals extend the idea of integration from intervals to regions in $\mathbb{R}^2$ and $\mathbb{R}^3$. They allow us to compute quantities such as areas, volumes, and other accumulated quantities over two- and three-dimensional regions. In this section, we develop the basic ideas and techniques for evaluating double and triple integrals.

3.1 Double integrals

In single-variable calculus, the definite integral of a nonnegative function represents the area under its graph. We now extend this idea to functions of two variables. For a nonnegative function $f(x, y)$, the double integral will represent the volume under the surface $z = f(x, y)$ over a region in the xy-plane.

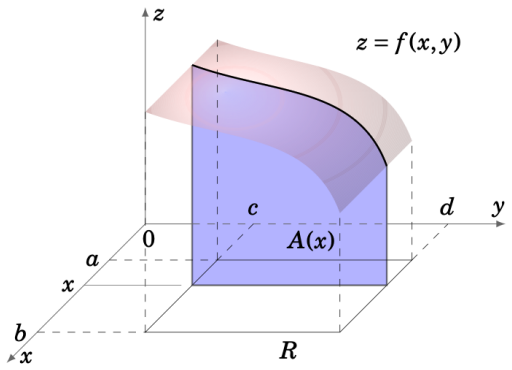
Let $f(x, y)$ be a continuous function such that $f(x, y) \geq 0$ on the rectangle

$$R = [a, b] \times [c, d].$$

Consider the surface $z = f(x, y)$ above the rectangle R. Fix a value $x \in [a, b]$ and intersect the surface with the vertical plane $x =$ constant. This gives the curve $z = f(x, y)$, where x is fixed and y varies from c to d. The area between this curve and the xy-plane is therefore

$$A(x) = \int_c^d f(x, y)\, dy.$$

Thus, for each fixed x, $A(x)$ gives the area of a vertical cross-section of the solid under the surface, as shown in the figure below.



Now let x vary from a to b. The total volume can be obtained by adding these cross-sectional areas. Hence, we integrate $A(x)$ with respect to x:

$$V = \int_a^b A(x)\,dx = \int_a^b \left(\int_c^d f(x,y)\,dy \right) dx.$$

This motivates the definition of the **double integral** of f over the rectangle R:

$$\iint_R f(x,y)\,dA = \int_a^b \int_c^d f(x,y)\,dy\,dx.$$

For a nonnegative function, this double integral represents the **volume under the surface** $z = f(x,y)$ and above the region R. The double integral above is an example of an **iterated integral**. The inner integral is evaluated first, treating the other variable as a constant, and the resulting function is then integrated with respect to the outer variable. Thus,

$$\int_a^b \int_c^d f(x,y)\,dy\,dx$$

means that we first integrate with respect to y and then with respect to x.

We could also take cross-sections parallel to the xz-plane and integrate in the opposite order. Thus,

$$\int_a^b \int_c^d f(x,y)\,dy\,dx = \int_c^d \int_a^b f(x,y)\,dx\,dy.$$

For a continuous function on a rectangle, this follows from **Fubini's theorem**. Therefore, the order of integration can be changed whenever it is convenient.

When writing an iterated integral, the differential next to the function indicates the variable of the inner integration. For example, in

$$\int_c^d \int_a^b f(x,y)\,dx\,dy,$$

we first integrate with respect to x and then with respect to y.

Example 21. *Find the volume under the surface $z = x + y$ over the rectangle $R = [0, 2] \times [0, 1]$.*
Solution: *Since $x + y \geq 0$ on R, the required volume is*

$$V = \int_0^1 \int_0^2 (x + y)\, dx\, dy.$$

Evaluating the inner integral first,

$$V = \int_0^1 \left[\frac{x^2}{2} + xy\right]_{x=0}^{x=2} dy = \int_0^1 (2 + 2y)\, dy = 3.$$

Changing the order of integration gives

$$V = \int_0^2 \int_0^1 (x + y)\, dy\, dx = \int_0^2 \left[xy + \frac{y^2}{2}\right]_{y=0}^{y=1} dx = \int_0^2 \left(x + \frac{1}{2}\right) dx = 3.$$

Thus, both orders of integration give the same volume.

Example 22. *Find the volume V under the surface $z = e^{x+y}$ over the rectangle $R = [2, 3] \times [1, 2]$.*
Solution: *Since $e^{x+y} > 0$ for all (x, y), we have*

$$V = \int_1^2 \int_2^3 e^{x+y}\, dx\, dy = \int_1^2 \left[e^{x+y}\right]_{x=2}^{x=3} dy = \int_1^2 \left(e^{y+3} - e^{y+2}\right) dy$$

$$= \left[e^{y+3} - e^{y+2}\right]_{y=1}^{y=2} = e^5 - e^4 - (e^4 - e^3) = e^5 - 2e^4 + e^3.$$

Recall that for a general function $f(x)$, the integral $\int_a^b f(x)\, dx$ represents the difference between the area below the curve $y = f(x)$ and above the x-axis when $f(x) \geq 0$, and the area above the curve and below the x-axis when $f(x) \leq 0$. Similarly, the double integral of a continuous function $f(x, y)$ represents the difference between the volume below the surface $z = f(x, y)$ and above the xy-plane when $f(x, y) \geq 0$, and the volume above the surface and below the xy-plane when $f(x, y) \leq 0$. Thus, double integrals can be evaluated using iterated integrals over a rectangle, regardless of whether $f(x, y)$ is positive or negative.

Example 23. *Evaluate*

$$\int_0^{\pi/2} \int_0^1 xy \cos(x^2 y)\, dx\, dy.$$

Since

$$\frac{\partial}{\partial x} \sin(x^2 y) = 2xy \cos(x^2 y),$$

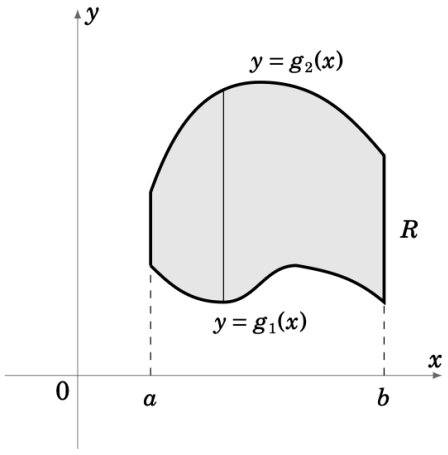
we have

$$\begin{aligned}\int_0^{\pi/2}\int_0^1 xy\cos(x^2y)\,dx\,dy &= \frac{1}{2}\int_0^{\pi/2}\left[\sin(x^2y)\right]_{x=0}^{x=1}dy\\ &= \frac{1}{2}\int_0^{\pi/2}\sin y\,dy\\ &= \frac{1}{2}.\end{aligned}$$

3.2 Double integrals over a general region

In the previous section we got an idea of what a double integral over a rectangle represents. We can now define the double integral of a real-valued function $f(x,y)$ over more general regions in $\mathbb{R}^2$.

Suppose that we have a region R in the xy-plane that is bounded on the left by the vertical line $x=a$, bounded on the right by the vertical line $x=b$ (where $a<b$), bounded below by a curve $y=g_1(x)$, and bounded above by a curve $y=g_2(x)$, as in Figure (a). We will assume that $g_1(x)$ and $g_2(x)$ do not intersect on the open interval (a,b) (they could intersect at the endpoints $x=a$ and $x=b$, though).

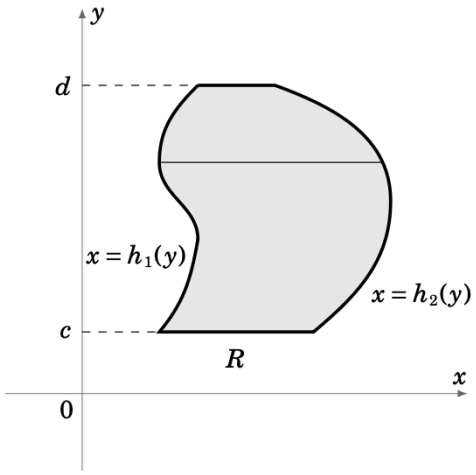


(a) Vertical slice: $\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$

Using the slice method from the previous section, the double integral of a real-valued function $f(x,y)$ over the region R, denoted by $\iint_R f(x,y)\,dA$, is given by

$$\iint_R f(x,y)\,dA = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right]dx.$$

Thus, we take vertical slices of the region R between the curves $y = g_1(x)$ and $y = g_2(x)$. The symbol dA is called an *area element*, with A denoting area. Notice that $f(x, y)$ is first integrated with respect to y, treating x as a constant. The result is then a function of x, which is integrated with respect to x.



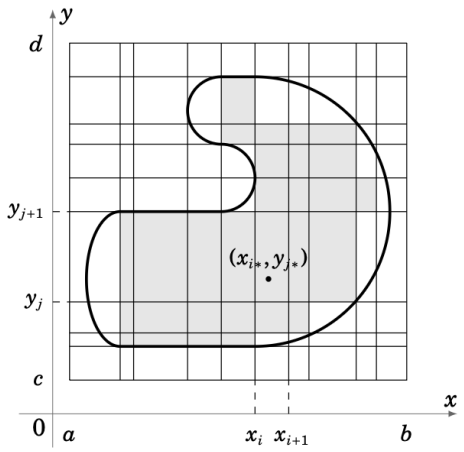
(b) Horizontal slice: $\int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$

Similarly, suppose that the region R is bounded on the left by the curve $x = h_1(y)$, bounded on the right by the curve $x = h_2(y)$, bounded below by the horizontal line $y = c$, and bounded above by the horizontal line $y = d$, where $c < d$, as in Figure (b). Assuming that $h_1(y)$ and $h_2(y)$ do not intersect on the open interval (c, d), taking horizontal slices gives

$$\iint_R f(x, y)\, dA = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx \right] dy.$$

Here, we first integrate with respect to x and then with respect to y. Thus, depending on the shape of the region, we may use either vertical or horizontal slices.

For a general region R, which may not be one of the types of regions considered so far, the double integral $\iint_R f(x, y)\, dA$ can be defined using a limiting process. Assume that $f(x, y)$ is a nonnegative real-valued function and that R is a bounded region in $\mathbb{R}^2$. Thus, R can be enclosed in some rectangle $[a, b] \times [c, d]$. Divide this rectangle into a grid of smaller subrectangles and consider only those subrectangles that are completely contained in R, as shown in the figure below



In each such subrectangle $[x_i, x_{i+1}] \times [y_j, y_{j+1}]$, choose a point (x_{i*}, y_{j*}). The volume under the surface $z = f(x, y)$ over this subrectangle is approximately

$$f(x_{i*}, y_{j*})\Delta x_i \Delta y_j,$$

where

$$\Delta x_i = x_{i+1} - x_i, \qquad \Delta y_j = y_{j+1} - y_j.$$

Here, $f(x_{i*}, y_{j*})$ represents the height of the surface, while $\Delta x_i \Delta y_j$ is the area of the corresponding subrectangle.

The total volume under the surface is therefore approximately the sum of the volumes over all such subrectangles:

$$\sum_i \sum_j f(x_{i*}, y_{j*})\Delta x_i \Delta y_j.$$

The summation is taken over the indices of the subrectangles that lie completely inside R.

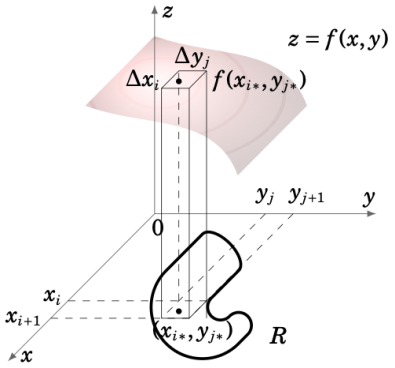
As the subrectangles are made smaller and smaller, the length of the largest diagonal of the subrectangles approaches 0. At the same time, the union of these subrectangles fills more and more of the region R. Hence, the above sum approaches the actual volume under the surface $z = f(x, y)$ over R.

The double integral of f over R is defined as this limiting value:

$$\iint_R f(x, y)\, dA = \lim \sum_i \sum_j f(x_{i*}, y_{j*})\Delta x_i \Delta y_j,$$

where the limit is taken as the length of the largest diagonal of the subrectangles tends to 0.

Over a small subrectangle, the surface is approximated by a rectangular box (parallelepiped) whose base area is $\Delta x_i \Delta y_j$ and whose height is $f(x_{i*}, y_{j*})$.



For a function $f(x, y)$ that is not necessarily nonnegative, the same definition applies. In this case, the integral represents a *signed volume*: regions where $f(x, y) > 0$ contribute positively, while regions where $f(x, y) < 0$ contribute negatively. Thus, the double integral can be defined for a continuous function f regardless of whether f is always positive or always negative.

For regions of the type considered earlier, the definition above agrees with the iterated-integral description. In particular, when

$$R = \{(x, y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

we have

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

Similarly, if

$$R = \{(x, y) : c \le y \le d,\ h_1(y) \le x \le h_2(y)\},$$

then

$$\iint_R f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy.$$

Finally, the region R need not be bounded. Double integrals over unbounded regions, or over regions where the function is unbounded, are called *improper double integrals*. They are defined using appropriate limits, in the same spirit as improper integrals of single-variable functions.

Lecture 6, October 26, 2026

Example 24. *Find the volume under the plane $z = 8x + 6y$ over the region*

$$R = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 2x^2\}.$$

The region R is shown in the figure below. Using vertical slices, we have

$$V = \iint_R (8x + 6y)\, dA = \int_0^1 \left[\int_0^{2x^2} (8x + 6y)\, dy \right] dx.$$

Thus,

$$\begin{aligned} V &= \int_0^1 (16x^3 + 12x^4)\, dx \\ &= \left[4x^4 + \frac{12}{5}x^5 \right]_0^1 = \frac{32}{5}. \end{aligned}$$

We can also describe the same region using horizontal slices. Since $y = 2x^2$ gives $x = \sqrt{y/2}$, we have

$$0 \le y \le 2, \qquad \sqrt{\frac{y}{2}} \le x \le 1.$$

Hence,

$$\begin{aligned} V &= \int_0^2 \left[\int_{\sqrt{y/2}}^1 (8x + 6y)\, dx \right] dy \\ &= \int_0^2 \left(4 + 4y - 3\sqrt{2}\, y^{3/2} \right) dy \\ &= \frac{32}{5}. \end{aligned}$$

Thus, both choices of slices give the same volume. The more convenient description of the region can therefore be chosen depending on the problem.

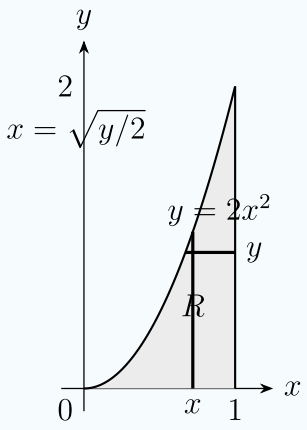
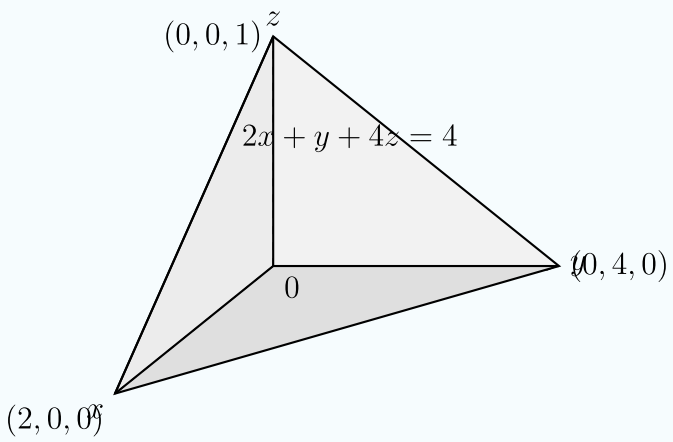
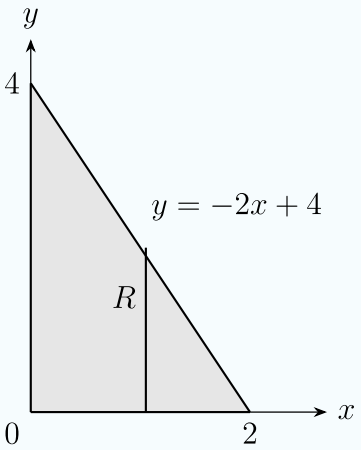


Figure 14: The region R with vertical and horizontal slices.

Example 25. *Find the volume V of the solid bounded by the three coordinate planes and the plane $2x + y + 4z = 4$.*





The solid is shown in the above figure, and its projection onto the xy-plane is shown above. Solving the equation of the plane for z, we obtain

$$z = \frac{1}{4}(4 - 2x - y).$$

Thus, the volume is

$$V = \iint_R \frac{1}{4}(4 - 2x - y)\, dA,$$

where

$$R = \{(x, y) : 0 \le x \le 2,\ 0 \le y \le -2x + 4\}.$$

Using vertical slices in R, we get

$$\begin{aligned} V &= \int_0^2 \left[\int_0^{-2x+4} \frac{1}{4}(4 - 2x - y)\, dy \right] dx \\ &= \int_0^2 \frac{1}{8}(4 - 2x)^2\, dx \\ &= -\frac{1}{48}(4 - 2x)^3 \Big|_0^2 \\ &= \frac{64}{48} = \frac{4}{3}. \end{aligned}$$

Example 26. *Evaluate*

$$\int_0^\infty \int_0^\infty xye^{-(x^2+y^2)}\, dx\, dy.$$

Solution: *Since*

$$e^{-(x^2+y^2)} = e^{-x^2}e^{-y^2},$$

we can separate the integral:

$$\begin{aligned}\int_0^\infty \int_0^\infty xye^{-(x^2+y^2)}\,dx\,dy &= \left(\int_0^\infty xe^{-x^2}\,dx\right)\left(\int_0^\infty ye^{-y^2}\,dy\right)\\ &= \left[-\frac{1}{2}e^{-x^2}\right]_0^\infty \left[-\frac{1}{2}e^{-y^2}\right]_0^\infty\\ &= \frac{1}{2}\cdot\frac{1}{2} = \boxed{\frac{1}{4}}.\end{aligned}$$

Example 27. *Evaluate*

$$\int_0^{\ln 2}\int_0^{\ln 5} e^{2x+y}\,dx\,dy.$$

Solution: *Since*

$$e^{2x+y} = e^{2x}e^y,$$

we have

$$\begin{aligned}\int_0^{\ln 2}\int_0^{\ln 5} e^{2x+y}\,dx\,dy &= \int_0^{\ln 2}\left[\frac{1}{2}e^{2x}\right]_0^{\ln 5} dy\\ &= \int_0^{\ln 2} 12e^y\,dy = 12[e^y]_0^{\ln 2} = 12.\end{aligned}$$

Example 28. *Find the volume bounded above by the paraboloid*

$$z = x^2 + y^2$$

and below by the square

$$R = \{(x,y) : -1 \le x \le 1,\ -1 \le y \le 1\}.$$

Solution: *The volume is given by*

$$\begin{aligned}V &= \iint_R (x^2+y^2)\,dA\\ &= \int_{-1}^1\int_{-1}^1 (x^2+y^2)\,dy\,dx\\ &= \int_{-1}^1 \left[x^2y + \frac{y^3}{3}\right]_{-1}^1 dx\\ &= \int_{-1}^1 \left(2x^2 + \frac{2}{3}\right)dx = \frac{8}{3}.\end{aligned}$$

Example 29. *Evaluate*

$$\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6x \, dy \, dx$$

by changing the order of integration.
Solution: *The region is the right half of the disk*

$$x^2 + y^2 \leq 4.$$

After changing the order of integration, we have

$$-2 \leq y \leq 2, \qquad 0 \leq x \leq \sqrt{4 - y^2}.$$

Hence,

$$\begin{aligned}\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6x \, dy \, dx &= \int_{-2}^2 \int_0^{\sqrt{4-y^2}} 6x \, dx \, dy \\ &= \int_{-2}^2 \left[3x^2\right]_0^{\sqrt{4-y^2}} dy \\ &= \int_{-2}^2 (12 - 3y^2) \, dy = 32.\end{aligned}$$

Example 30. *Find the area of the region bounded by the parabolas*

$$y^2 = 4ax \qquad \text{and} \qquad x^2 = 4ay,$$

where $a > 0$.
Solution: *The curves intersect when*

$$y^2 = 4ax, \qquad x^2 = 4ay.$$

For the enclosed region, we have $x, y \geq 0$. *Substituting* $y = x$ *gives*

$$x^2 = 4ax,$$

so the points of intersection are $(0, 0)$ *and* $(4a, 4a)$.
Writing the curves as functions of x,

$$y = 2\sqrt{ax} \qquad \text{and} \qquad y = \frac{x^2}{4a}.$$

Thus, the required area is

$$\begin{aligned}A &= \int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a}\right) dx \\ &= \left[\frac{4}{3}\sqrt{a}\, x^{3/2} - \frac{x^3}{12a}\right]_0^{4a} \\ &= \frac{32}{3}a^2 - \frac{16}{3}a^2 = \frac{16}{3}a^2.\end{aligned}$$

3.3 Triple integrals

Our definition of a double integral of a real-valued function $f(x, y)$ over a region R in $\mathbb{R}^2$ can be extended to define a *triple integral* of a real-valued function $f(x, y, z)$ over a solid S in $\mathbb{R}^3$. We proceed in the same way as before. The solid S can be enclosed in some rectangular parallelepiped, which is then divided into smaller subparallelepipeds. In each subparallelepiped inside S, with side lengths Δx, Δy, and Δz, choose a point (x_*, y_*, z_*). The triple integral of $f(x, y, z)$ over S, denoted by $\iiint_S f(x, y, z)\, dV$, is defined by

$$\iiint_S f(x, y, z)\, dV = \lim \sum_i \sum_j \sum_k f(x_*, y_*, z_*)\, \Delta x\, \Delta y\, \Delta z.$$

Here, the limit is taken over all subdivisions of the rectangular parallelepiped enclosing S into subparallelepipeds whose largest diagonal tends to 0, and the triple sum is taken over all subparallelepipeds contained in S. It can be shown that this limit does not depend on the choice of the rectangular parallelepiped containing S. The symbol dV is called the ***volume element***.

Physically, the triple integral can be viewed as a natural extension of the double integral. Just as a double integral can represent the volume under a surface, a triple integral represents the ***hypervolume*** under a three-dimensional hypersurface

$$w = f(x, y, z),$$

whose graph lies in $\mathbb{R}^4$. In general, the term "volume" is used for the corresponding notion in any dimension; for example, length in $\mathbb{R}$ and area in $\mathbb{R}^2$.

In the simplest case, suppose that S is a rectangular parallelepiped

$$S = [x_1, x_2] \times [y_1, y_2] \times [z_1, z_2].$$

Equivalently,

$$S = \{(x, y, z) : x_1 \le x \le x_2,\ y_1 \le y \le y_2,\ z_1 \le z \le z_2\}.$$

Then the triple integral can be evaluated as a sequence of three iterated integrals:

$$\iiint_S f(x, y, z)\, dV = \int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y, z)\, dx\, dy\, dz.$$

For a rectangular parallelepiped, the order of integration can be changed, provided the corresponding limits are adjusted accordingly.

A more general situation occurs when S is bounded below by the surface

$$z = g_1(x, y),$$

and above by the surface

$$z = g_2(x, y),$$

while y lies between the curves

$$y = h_1(x), \qquad y = h_2(x),$$

and x varies between a and b. In this case,

$$\iiint_S f(x,y,z)\,dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)\,dz\,dy\,dx.$$

Thus, as with double integrals, the triple integral is evaluated by integrating successively with respect to one variable at a time, with the limits determined by the geometry of the solid.

Example 31. *Evaluate*

$$\int_0^2 \int_0^1 \int_0^3 (2x+y+z)\,dz\,dx\,dy.$$

Solution:

$$\begin{aligned}
\int_0^2 \int_0^1 \int_0^3 (2x+y+z)\,dz\,dx\,dy &= \int_0^2 \int_0^1 \left(2xz + yz + \frac{z^2}{2}\right)\Bigg|_{z=0}^{z=3} dx\,dy \\
&= \int_0^2 \int_0^1 \left(6x + 3y + \frac{9}{2}\right) dx\,dy \\
&= \int_0^2 \left(3x^2 + 3xy + \frac{9}{2}x\right)\Bigg|_{x=0}^{x=1} dy \\
&= \int_0^2 \left(\frac{15}{2} + 3y\right) dy \\
&= \left(\frac{15}{2}y + \frac{3}{2}y^2\right)\Bigg|_0^2 = 18.
\end{aligned}$$

Example 32. *Evaluate*

$$\int_0^1 \int_0^{1-x} \int_0^{2-x-y} (x+y+z)\,dz\,dy\,dx.$$

Solution:

$$\begin{aligned}
&\int_0^1 \int_0^{1-x} \int_0^{2-x-y} (x+y+z)\,dz\,dy\,dx \\
&= \int_0^1 \int_0^{1-x} \left((x+y)z + \frac{1}{2}z^2\right)\Big|_{z=0}^{z=2-x-y} dy\,dx \\
&= \int_0^1 \int_0^{1-x} \left((x+y)(2-x-y) + \frac{1}{2}(2-x-y)^2\right) dy\,dx \\
&= \int_0^1 \int_0^{1-x} \left(2 - \frac{x^2}{2} - xy - \frac{y^2}{2}\right) dy\,dx \\
&= \int_0^1 \left(2y - \frac{x^2}{2}y - \frac{x}{2}y^2 - \frac{1}{6}y^3\right)\Big|_{y=0}^{y=1-x} dx \\
&= \int_0^1 \left(\frac{11}{6} - 2x + \frac{1}{6}x^3\right) dx \\
&= \left(\frac{11}{6}x - x^2 + \frac{1}{24}x^4\right)\Big|_0^1 = \frac{7}{8}.
\end{aligned}$$

Note that the volume V of a solid in $\mathbb{R}^3$ is given by

$$V = \iiint_S 1\,dV.$$

Since the function being integrated is the constant 1, the above triple integral reduces to a double integral of the type considered in the previous section if the solid is bounded above by a surface $z = f(x,y)$ and bounded below by the xy-plane $z = 0$. There are many other possibilities. For example, the solid could be bounded below and above by surfaces $z = g_1(x,y)$ and $z = g_2(x,y)$, respectively, with y bounded between two curves $h_1(x)$ and $h_2(x)$, and x varying between a and b. Then

$$V = \iiint_S 1\,dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} 1\,dz\,dy\,dx = \int_a^b \int_{h_1(x)}^{h_2(x)} \big(g_2(x,y) - g_1(x,y)\big)\,dy\,dx.$$

Example 33. *Find the volume V of the solid S bounded by the three coordinate planes, the plane*

$$x + y + z = 2, \text{ and } z = x + y.$$

Solution: *The three coordinate planes are $x = 0$, $y = 0$, and $z = 0$, so the solid lies in the first octant. The two given planes can be written as*

$$z = x + y, \qquad\qquad z = 2 - x - y.$$

These two planes intersect when $x + y = 1$, with $z = 1$.
For $0 \le x + y \le 1$, the plane $z = x + y$ lies below $z = 2 - x - y$. Hence, the solid is bounded below by $z = 0$ and above by $z = x + y$.

For $1 \le x+y \le 2$, the plane $z = 2-x-y$ lies below $z = x+y$. Hence, the solid is bounded below by $z = 0$ and above by $z = 2-x-y$.
Therefore,

$$\begin{aligned} V &= \int_0^1 \int_0^{1-x} \int_0^{x+y} dz\, dy\, dx + \int_0^1 \int_{1-x}^{2-x} \int_0^{2-x-y} dz\, dy\, dx \\ &= \int_0^1 \int_0^{1-x} (x+y)\, dy\, dx + \int_0^1 \int_{1-x}^{2-x} (2-x-y)\, dy\, dx \\ &= \frac{1}{3} + 1 = \boxed{\frac{4}{3}}. \end{aligned}$$

Lecture 7, October 27, 2026

3.4 Change of variables in multiple integrals

Multiple integrals can often be simplified by making a suitable change of variables. This is similar to the substitution method used in single-variable calculus, although the procedure in several variables is slightly more involved. We now develop this method and see how an appropriate change of variables can simplify the evaluation of multiple integrals.

Recall the substitution method from single-variable calculus. For example, consider

$$\int_1^2 x^3 \sqrt{x^2-1}\, dx.$$

We make the substitution

$$u = x^2 - 1, \qquad du = 2x\, dx, \qquad x^2 = u + 1.$$

The limits change as follows:

x	1	2
$u = x^2 - 1$	0	3

Thus,

$$\begin{aligned} \int_1^2 x^3 \sqrt{x^2-1}\, dx &= \frac{1}{2} \int_0^3 (u+1)\sqrt{u}\, du \\ &= \frac{1}{2} \int_0^3 \left(u^{3/2} + u^{1/2}\right) du = \frac{14\sqrt{3}}{5}. \end{aligned}$$

Let us look more closely at what happened in this substitution. We set

$$u = x^2 - 1.$$

On the interval $[1, 2]$, the function $x \mapsto x^2 - 1$ is strictly increasing and maps $[1, 2]$ onto $[0, 3]$. Hence, it has an inverse function on $[0, 3]$. Thus, we can write x as a function of u:

$$x = g(u) = \sqrt{u + 1}.$$

Substituting $x = g(u) = \sqrt{u + 1}$ gives

$$f(x) = f(g(u)) = (u + 1)^{3/2}\sqrt{u},$$

and

$$dx = g'(u)\, du = \frac{1}{2}(u + 1)^{-1/2}\, du.$$

The limits also change according to

$$g(0) = 1, \qquad g(3) = 2.$$

Thus,

$$\int_1^2 f(x)\, dx = \int_0^3 f(g(u))g'(u)\, du.$$

More generally, let $x = g(u)$ be a one-to-one differentiable function from $[c, d]$ onto $[a, b]$, with $g'(u) \neq 0$ on $[c, d]$. Then

$$a = g(c), \qquad b = g(d),$$

and hence

$$c = g^{-1}(a), \qquad d = g^{-1}(b).$$

Therefore,

$$\boxed{\int_a^b f(x)\, dx = \int_{g^{-1}(a)}^{g^{-1}(b)} f(g(u))g'(u)\, du.}$$

This is the *change of variable formula* for single-variable integrals. The same idea will now be extended to multiple integrals, where two or more variables are changed simultaneously.

Theorem 12 (Change of Variables Formula for Multiple Integrals)**.** *Let*

$$x = x(u, v), \qquad y = y(u, v)$$

define a one-to-one mapping of a region R' in the uv-plane onto a region R in the xy-plane. Suppose that

$$J(u, v) = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} \neq 0$$

throughout R'. Then

$$\iint_R f(x,y)\,dA = \iint_{R'} f\big(x(u,v),y(u,v)\big)\,|J(u,v)|\,dA.$$

Similarly, let

$$x = x(u,v,w), \qquad y = y(u,v,w), \qquad z = z(u,v,w)$$

define a one-to-one mapping of a solid S' in the uvw-space onto a solid S in the xyz-space. Suppose that

$$J(u,v,w) = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix} \neq 0$$

throughout S'. Then

$$\iiint_S f(x,y,z)\,dV = \iiint_{S'} f\big(x(u,v,w),y(u,v,w),z(u,v,w)\big)\,|J(u,v,w)|\,dV.$$

The determinant $J(u,v)$ is called the ***Jacobian*** of x and y with respect to u and v. It is also denoted by

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)}.$$

Similarly, for three variables,

$$J(u,v,w) = \frac{\partial(x,y,z)}{\partial(u,v,w)}.$$

The change of variables formula can be viewed as a multivariable version of the single-variable relation

$$dx = g'(u)\,du.$$

In two variables,

$$dA(x,y) = |J(u,v)|\,dA(u,v),$$

while in three variables,

$$dV(x,y,z) = |J(u,v,w)|\,dV(u,v,w).$$

Example 34. *Evaluate*

$$\iint_R \frac{x-y}{e^{x+y}}\,dA, \qquad R = \{(x,y) : x \geq 0,\ y \geq 0,\ x+y \leq 1\}.$$

Solution: *The integral is not convenient to evaluate directly. Since the numerator and denominator involve $x-y$ and $x+y$, respectively, we choose the change of variables*

$$u = x - y, \qquad v = x + y.$$

Solving for x and y in terms of u and v, we obtain

$$x = \frac{u+v}{2}, \qquad y = \frac{v-u}{2}.$$

The original region R is the triangle bounded by $x = 0$, $y = 0$, and $x + y = 1$. Under the above change of variables,

$$x \geq 0 \quad \Longleftrightarrow \quad u + v \geq 0,$$

$$y \geq 0 \quad \Longleftrightarrow \quad v - u \geq 0,$$

and

$$x + y \leq 1 \quad \Longleftrightarrow \quad v \leq 1.$$

Hence, the corresponding region R' in the uv-plane is

$$0 \leq v \leq 1, \qquad -v \leq u \leq v.$$

The regions R and R' are shown below.

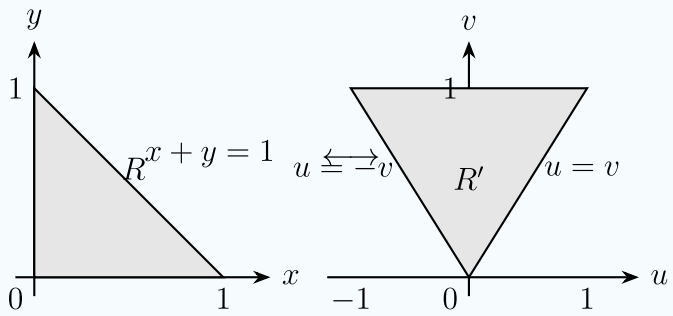


Figure 15: The regions R and R'.

The Jacobian is

$$J(u,v) = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \dfrac{1}{2} & \dfrac{1}{2} \\ -\dfrac{1}{2} & \dfrac{1}{2} \end{vmatrix} = \frac{1}{2}.$$

Hence,

$$|J(u,v)| = \frac{1}{2}.$$

Also,

$$x - y = u, \qquad x + y = v.$$

Therefore,

$$\begin{aligned}\iint_R e^{\frac{x-y}{x+y}}\,dA &= \iint_{R'} e^{u/v}|J(u,v)|\,dA \\ &= \frac{1}{2}\int_0^1 \int_{-v}^{v} e^{u/v}\,du\,dv \\ &= \frac{1}{2}\int_0^1 \left[ve^{u/v}\right]_{u=-v}^{u=v}\,dv \\ &= \frac{1}{2}\int_0^1 v(e-e^{-1})\,dv \\ &= \frac{e-e^{-1}}{4}.\end{aligned}$$

The change of variables formula can also be used to evaluate double integrals in polar coordinates. Recall that

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The corresponding Jacobian is

$$J(r,\theta) = \begin{vmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

Since $r \geq 0$, we have

$$|J(r,\theta)| = r.$$

Therefore,

$$\boxed{\iint_R f(x,y)\,dA = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dA.}$$

Here, R' is the corresponding region in the $r\theta$-plane under the transformation

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Example 35. *Find the volume V inside the paraboloid*

$$z = x^2 + y^2$$

for $0 \leq z \leq 1$.
Solution: *The solid lies between the paraboloid $z = x^2 + y^2$ and the plane $z = 1$. Its projection onto the xy-plane is*

$$R = \{(x,y) : x^2 + y^2 \leq 1\},$$

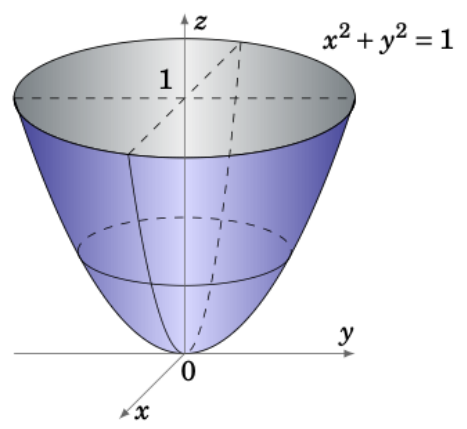
which is the unit disk.
Using vertical slices, the height of the solid above a point $(x, y) \in R$ *is*

$$1 - (x^2 + y^2).$$

Hence,

$$V = \iint_R \left(1 - x^2 - y^2\right) dA.$$

The solid and its projection are shown below.



In polar coordinates,

$$x^2 + y^2 = r^2, \qquad dA = r\, dr\, d\theta,$$

and the unit disk becomes

$$0 \le r \le 1, \qquad 0 \le \theta \le 2\pi.$$

Therefore,

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^1 (1 - r^2) r\, dr\, d\theta \\ &= \int_0^{2\pi} \int_0^1 (r - r^3)\, dr\, d\theta \\ &= \int_0^{2\pi} \left[\frac{r^2}{2} - \frac{r^4}{4}\right]_0^1 d\theta = \int_0^{2\pi} \frac{1}{4}\, d\theta = \frac{\pi}{2}. \end{aligned}$$

Triple Integral in Cylindrical Coordinates

$$\iiint_S f(x, y, z)\, dx\, dy\, dz = \iiint_{S'} f(r\cos\theta, r\sin\theta, z)\, r\, dr\, d\theta\, dz,$$

where the mapping

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z$$

maps the solid S' in $r\theta z$-space onto the solid S in xyz-space in a one-to-one manner.

Triple Integral in Spherical Coordinates

$$\iiint_S f(x,y,z)\,dx\,dy\,dz = \iiint_{S'} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\,\rho^2\sin\phi\,d\rho\,d\phi\,d\theta,$$

where the mapping

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi$$

maps the solid S' in $\rho\phi\theta$-space onto the solid S in xyz-space in a one-to-one manner.

Example 36. *Find the volume V inside the sphere*

$$x^2 + y^2 + z^2 = a^2, \qquad a > 0,$$

and inside the cone

$$z \geq \sqrt{x^2 + y^2}.$$

Solution: *In spherical coordinates,*

$$x^2 + y^2 + z^2 = \rho^2, \qquad z = \rho\cos\phi, \qquad \sqrt{x^2 + y^2} = \rho\sin\phi.$$

Thus, the sphere is given by

$$0 \leq \rho \leq a,$$

while the cone gives

$$\rho\cos\phi \geq \rho\sin\phi \quad \Longrightarrow \quad 0 \leq \phi \leq \frac{\pi}{4}.$$

Also, $0 \le \theta \le 2\pi$. *Therefore,*

$$\begin{aligned} V &= \iiint_S 1\, dV \\ &= \int_0^{2\pi} \int_0^{\pi/4} \int_0^a \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/4} \frac{a^3}{3} \sin\phi \, d\phi \, d\theta \\ &= \int_0^{2\pi} \frac{a^3}{3} \left[-\cos\phi\right]_{\phi=0}^{\phi=\pi/4} d\theta \\ &= \int_0^{2\pi} \frac{a^3}{3} \left(1 - \frac{1}{\sqrt{2}}\right) d\theta \\ &= \boxed{\frac{2\pi a^3}{3}\left(1 - \frac{1}{\sqrt{2}}\right)}. \end{aligned}$$

Example 37. *Show that the* Beta function, *defined by*

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\, dt, \qquad x > 0,\ y > 0,$$

satisfies

$$B(y,x) = B(x,y).$$

Solution: *Starting from* $B(y,x)$, *we have*

$$B(y,x) = \int_0^1 t^{y-1}(1-t)^{x-1}\, dt.$$

Make the substitution

$$t = 1 - u, \qquad dt = -du.$$

The limits change as

t	0	1
$u = 1 - t$	1	0

Therefore,

$$\begin{aligned} B(y,x) &= \int_1^0 (1-u)^{y-1} u^{x-1}(-du) \\ &= \int_0^1 u^{x-1}(1-u)^{y-1}\, du \\ &= B(x,y). \end{aligned}$$

Hence,

$$\boxed{B(y,x) = B(x,y)}.$$

Example 38. *Fix $a > 0$. Show that*

(i)

$$\int_{\mathbb{R}^2} e^{-a(x^2+y^2)}\,dx\,dy = \frac{\pi}{a},$$

(ii)

$$\int_{\mathbb{R}} e^{-ax^2}\,dx = \sqrt{\frac{\pi}{a}},$$

(iii)

$$\int_{\mathbb{R}^3} e^{-a(x^2+y^2+z^2)}\,dx\,dy\,dz = \left(\frac{\pi}{a}\right)^{3/2}.$$

Solution.
(i) *We use polar coordinates*

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

for which

$$x^2 + y^2 = r^2, \qquad dx\,dy = r\,dr\,d\theta.$$

Since $(x, y) \in \mathbb{R}^2$ corresponds to $0 \le r < \infty$ and $0 \le \theta \le 2\pi$, we obtain

$$\begin{aligned}\int_{\mathbb{R}^2} e^{-a(x^2+y^2)}\,dx\,dy &= \int_0^{2\pi}\int_0^{\infty} e^{-ar^2} r\,dr\,d\theta \\ &= \int_0^{2\pi}\left[-\frac{1}{2a}e^{-ar^2}\right]_0^{\infty} d\theta \\ &= \frac{1}{2a}\int_0^{2\pi} d\theta \\ &= \frac{\pi}{a}.\end{aligned}$$

(ii) *Let*

$$I = \int_{\mathbb{R}} e^{-ax^2}\,dx.$$

Then

$$\begin{aligned}I^2 &= \left(\int_{\mathbb{R}} e^{-ax^2}\,dx\right)\left(\int_{\mathbb{R}} e^{-ay^2}\,dy\right) \\ &= \int_{\mathbb{R}^2} e^{-a(x^2+y^2)}\,dx\,dy \\ &= \frac{\pi}{a}.\end{aligned}$$

Since $I > 0$, we obtain

$$\boxed{\int_{\mathbb{R}} e^{-ax^2}\,dx = \sqrt{\frac{\pi}{a}}.}$$

(iii) *We now consider the corresponding integral in* $\mathbb{R}^3$*:*

$$I_3 = \int_{\mathbb{R}^3} e^{-a(x^2+y^2+z^2)}\, dx\, dy\, dz.$$

Using cylindrical coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$

with

$$x^2 + y^2 = r^2, \qquad dx\, dy\, dz = r\, dr\, d\theta\, dz.$$

Therefore,

$$\begin{aligned} I_3 &= \int_{-\infty}^{\infty}\int_0^{2\pi}\int_0^{\infty} e^{-a(r^2+z^2)} r\, dr\, d\theta\, dz \\ &= \int_{-\infty}^{\infty} e^{-az^2}\left(\int_0^{2\pi}\int_0^{\infty} e^{-ar^2} r\, dr\, d\theta\right) dz \\ &= \frac{\pi}{a}\int_{-\infty}^{\infty} e^{-az^2}\, dz. \end{aligned}$$

Using part (ii),

$$I_3 = \frac{\pi}{a}\sqrt{\frac{\pi}{a}} = \left(\frac{\pi}{a}\right)^{3/2}.$$

3.5 Application: center of mass

Before deriving the formula for the center of mass of a region, let us first recall the case of finitely many point masses.

Suppose that masses $m_1, m_2, \ldots, m_n$ are placed at the points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ in the plane. The total mass is

$$M = m_1 + m_2 + \cdots + m_n.$$

The center of mass $(\bar{x}, \bar{y})$ is the point whose coordinates are the mass-weighted averages of the corresponding coordinates of the masses. Thus,

$$\begin{aligned} \bar{x} &= \frac{x_1 m_1 + x_2 m_2 + \cdots + x_n m_n}{M}, \\ \bar{y} &= \frac{y_1 m_1 + y_2 m_2 + \cdots + y_n m_n}{M}. \end{aligned}$$

The quantities

$$M_y = x_1 m_1 + x_2 m_2 + \cdots + x_n m_n,$$
$$M_x = y_1 m_1 + y_2 m_2 + \cdots + y_n m_n$$

are called the **moments** of the system about the y-axis and x-axis, respectively. Notice that the moment about the y-axis involves the distance x_i from the y-axis, while the moment about the x-axis involves the distance y_i from the x-axis.

Hence,

$$\boxed{\bar{x} = \frac{M_y}{M}}, \qquad \boxed{\bar{y} = \frac{M_x}{M}}.$$

We now extend this idea to a continuous region R. Divide R into a large number of small rectangles. Consider one such rectangle with area ΔA containing a point (x_i, y_i). If the density of the region is $\delta(x, y)$, then the mass of this small rectangle is approximately

$$\Delta m_i \approx \delta(x_i, y_i)\Delta A.$$

Therefore, its contribution to the moment about the y-axis is

$$x_i \Delta m_i \approx x_i \delta(x_i, y_i)\Delta A.$$

Adding the contributions from all the small rectangles and taking the limit as their sizes approach zero gives

$$M_y = \iint_R x\delta(x, y)\, dA.$$

Similarly, the moment about the x-axis is

$$M_x = \iint_R y\delta(x, y)\, dA.$$

The total mass is obtained by adding the masses of all the small rectangles:

$$M = \iint_R \delta(x, y)\, dA.$$

Consequently, the center of mass $(\bar{x}, \bar{y})$ of the region R is given by

$$\boxed{\bar{x} = \frac{M_y}{M} = \frac{\displaystyle\iint_R x\delta(x, y)\, dA}{\displaystyle\iint_R \delta(x, y)\, dA}}$$

and

$$\boxed{\bar{y} = \frac{M_x}{M} = \frac{\displaystyle\iint_R y\delta(x,y)\,dA}{\displaystyle\iint_R \delta(x,y)\,dA}.}$$

In particular, if the density is uniform, then $\delta(x,y) = 1$ (we may take the constant density to be 1). In this case,

$$M = \iint_R dA,$$
$$M_y = \iint_R x\,dA,$$
$$M_x = \iint_R y\,dA.$$

Thus, for a region of uniform density,

$$\boxed{\bar{x} = \frac{\displaystyle\iint_R x\,dA}{\displaystyle\iint_R dA}, \qquad \bar{y} = \frac{\displaystyle\iint_R y\,dA}{\displaystyle\iint_R dA}.}$$

So, the center of mass can be viewed as a **mass-weighted average of position**. The moments measure how the mass is distributed relative to the coordinate axes, while dividing by the total mass gives the corresponding average position.

The same idea extends naturally from a region in $\mathbb{R}^2$ to a solid in $\mathbb{R}^3$. Let S be a solid with continuous mass density $\delta(x,y,z)$ at a point $(x,y,z) \in S$. We divide S into small rectangular boxes. If a small box has volume ΔV and contains the point (x_i, y_i, z_i), then its mass is approximately

$$\Delta m_i \approx \delta(x_i, y_i, z_i)\Delta V.$$

For the center of mass $(\bar{x}, \bar{y}, \bar{z})$, each coordinate is again the corresponding mass-weighted average. Thus,

$$\bar{x} = \frac{\sum x_i \Delta m_i}{\sum \Delta m_i},$$
$$\bar{y} = \frac{\sum y_i \Delta m_i}{\sum \Delta m_i},$$
$$\bar{z} = \frac{\sum z_i \Delta m_i}{\sum \Delta m_i}.$$

Passing to the limit, the total mass of S is

$$M = \iiint_S \delta(x, y, z)\, dV.$$

The moment about the yz-plane is obtained by multiplying each small mass by its distance x from the yz-plane. Hence,

$$M_{yz} = \iiint_S x\delta(x, y, z)\, dV.$$

Similarly, the distance of (x, y, z) from the xz-plane is y, and its distance from the xy-plane is z. Therefore,

$$M_{xz} = \iiint_S y\delta(x, y, z)\, dV,$$
$$M_{xy} = \iiint_S z\delta(x, y, z)\, dV.$$

Consequently, the coordinates of the center of mass are

$$\boxed{\bar{x} = \frac{M_{yz}}{M}, \qquad \bar{y} = \frac{M_{xz}}{M}, \qquad \bar{z} = \frac{M_{xy}}{M}.}$$

Equivalently,

$$\bar{x} = \frac{\displaystyle\iiint_S x\delta(x, y, z)\, dV}{\displaystyle\iiint_S \delta(x, y, z)\, dV},$$
$$\bar{y} = \frac{\displaystyle\iiint_S y\delta(x, y, z)\, dV}{\displaystyle\iiint_S \delta(x, y, z)\, dV},$$
$$\bar{z} = \frac{\displaystyle\iiint_S z\delta(x, y, z)\, dV}{\displaystyle\iiint_S \delta(x, y, z)\, dV}.$$

The important point to remember is that the moment about a coordinate plane uses the **distance from that plane**. Thus,

Plane	**Distance**	**Moment**
yz-plane	x	M_{yz}
xz-plane	y	M_{xz}
xy-plane	z	M_{xy}

Therefore, just as in the two-dimensional case, the center of mass is the mass-weighted average of the position of the points in the solid.

Example 39. *Find the center of mass of the region*

$$R = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 2x^2\},$$

if the density function at (x, y) *is*

$$\delta(x, y) = x + y.$$

Solution: *The region* R *is shown in Figure 16. We first find the total mass:*

$$\begin{aligned} M &= \iint_R \delta(x, y)\, dA \\ &= \int_0^1 \int_0^{2x^2} (x + y)\, dy\, dx \\ &= \int_0^1 \left(xy + \frac{y^2}{2} \right)_{y=0}^{y=2x^2} dx \\ &= \int_0^1 (2x^3 + 2x^4)\, dx \\ &= \left(\frac{x^4}{2} + \frac{2x^5}{5} \right)_0^1 = \frac{9}{10}. \end{aligned}$$

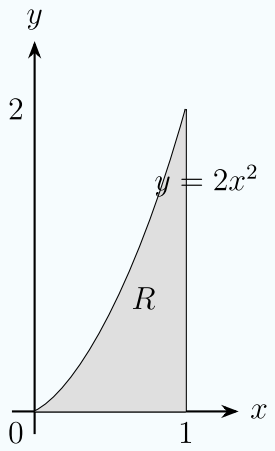


Figure 16: The region R.

The moments about the x- and y-axes are

$$\begin{aligned} M_x &= \iint_R y\delta(x,y)\,dA \\ &= \int_0^1 \int_0^{2x^2} y(x+y)\,dy\,dx \\ &= \int_0^1 \left(\frac{xy^2}{2} + \frac{y^3}{3}\right)_{y=0}^{y=2x^2} dx \\ &= \int_0^1 \left(2x^5 + \frac{8x^6}{3}\right) dx \\ &= \left(\frac{x^6}{3} + \frac{8x^7}{21}\right)_0^1 = \frac{5}{7}, \end{aligned}$$

and

$$\begin{aligned} M_y &= \iint_R x\delta(x,y)\,dA \\ &= \int_0^1 \int_0^{2x^2} x(x+y)\,dy\,dx \\ &= \int_0^1 \left(x^2y + \frac{xy^2}{2}\right)_{y=0}^{y=2x^2} dx \\ &= \int_0^1 (2x^4 + 2x^5)\,dx \\ &= \left(\frac{2x^5}{5} + \frac{x^6}{3}\right)_0^1 = \frac{11}{15}. \end{aligned}$$

Therefore, the center of mass $(\bar{x}, \bar{y})$ is given by

$$\bar{x} = \frac{M_y}{M} = \frac{11/15}{9/10} = \frac{22}{27}, \qquad \bar{y} = \frac{M_x}{M} = \frac{5/7}{9/10} = \frac{50}{63}.$$

Hence,

$$\boxed{(\bar{x}, \bar{y}) = \left(\frac{22}{27}, \frac{50}{63}\right)}.$$

Example 40. *Find the center of mass of the solid*

$$S = \{(x,y,z) : z \geq 0,\ x^2 + y^2 + z^2 \leq a^2\},$$

if the density function at (x,y,z) is $\delta(x,y,z) = 1$.
Solution: *The solid S is the upper hemisphere inside the sphere of radius a centered at the origin (see Figure 17). Since the density function is constant and S is symmetric*

about the z-axis, it is clear that

$$\bar{x} = 0, \qquad \bar{y} = 0.$$

We therefore need only find $\bar{z}$. The total mass is

$$\begin{aligned} M &= \iiint_S \delta(x, y, z)\, dV \\ &= \iiint_S 1\, dV = \text{Volume}(S). \end{aligned}$$

Since the volume of S is half the volume of the sphere of radius a, we have

$$M = \frac{1}{2}\left(\frac{4\pi a^3}{3}\right) = \frac{2\pi a^3}{3}.$$

The moment about the xy-plane is

$$\begin{aligned} M_{xy} &= \iiint_S z\delta(x, y, z)\, dV \\ &= \iiint_S z\, dV. \end{aligned}$$

Using spherical coordinates,

$$z = \rho\cos\phi, \qquad dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta.$$

For the upper hemisphere,

$$0 \le \rho \le a, \qquad 0 \le \phi \le \frac{\pi}{2}, \qquad 0 \le \theta \le 2\pi.$$

Hence,

$$\begin{aligned} M_{xy} &= \int_0^{2\pi}\int_0^{\pi/2}\int_0^a (\rho\cos\phi)\rho^2\sin\phi\, d\rho\, d\phi\, d\theta \\ &= \int_0^{2\pi}\int_0^{\pi/2} \sin\phi\cos\phi \left(\int_0^a \rho^3\, d\rho\right) d\phi\, d\theta \\ &= \int_0^{2\pi}\int_0^{\pi/2} \frac{a^4}{4}\sin\phi\cos\phi\, d\phi\, d\theta \\ &= \int_0^{2\pi}\int_0^{\pi/2} \frac{a^4}{8}\sin(2\phi)\, d\phi\, d\theta \\ &= \int_0^{2\pi} \left[-\frac{a^4}{16}\cos(2\phi)\right]_{\phi=0}^{\phi=\pi/2} d\theta \\ &= \int_0^{2\pi} \frac{a^4}{8}\, d\theta \\ &= \frac{\pi a^4}{4}. \end{aligned}$$

Therefore,

$$\bar{z} = \frac{M_{xy}}{M} = \frac{\pi a^4/4}{2\pi a^3/3} = \frac{3a}{8}.$$

Thus, the center of mass of S is

$$\boxed{(\bar{x}, \bar{y}, \bar{z}) = \left(0, 0, \frac{3a}{8}\right)}.$$

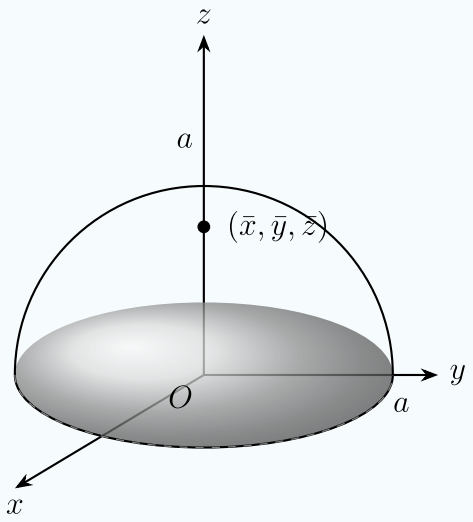


Figure 17: The upper hemisphere S.

Lecture 8, October 30, 2026

4 Line and surface integrals

In the previous sections, we studied multiple integrals over regions in $\mathbb{R}^2$ and solids in $\mathbb{R}^3$. We now extend the idea of integration to curves and surfaces. For example, we may wish to calculate the work done by a force along a prescribed path, or measure the flow of a vector field through a surface. Such problems lead to the notions of *line integrals* and *surface integrals*, respectively. These integrals provide important tools for studying vector fields and form the basis for some fundamental results of vector calculus, including Green's theorem, Stokes' theorem, and the divergence theorem.

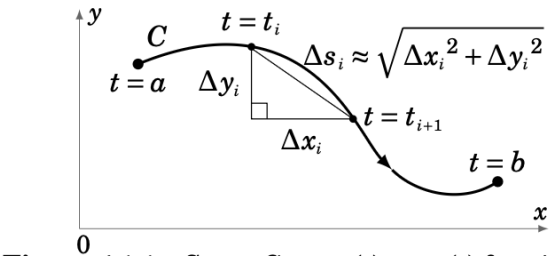
4.1 Line integrals

In single-variable calculus, the definite integral (usually called a Riemann integral) $\int_a^b f(x)\,dx$ can be used to calculate the work done by a force acting along the x-axis. When an object moves along a curve in $\mathbb{R}^2$ or $\mathbb{R}^3$, however, the force may vary with position and direction. We therefore need to extend the idea of integration to functions defined along curves.

In physics, the intuitive idea of work is that

$$\text{Work} = \text{Force} \times \text{Distance}.$$

Suppose that we want to find the total amount W of work done in moving an object along a curve C in $\mathbb{R}^2$ with a smooth parametrization $x = x(t)$, $y = y(t)$, $a \le t \le b$, with a force $f(x, y)$ which varies with the position (x, y) of the object and is applied in the direction of motion along C (see the figure below).



We assume that $f(x, y)$ is continuous and real-valued, so that we consider only the magnitude of the force. Partition the interval $[a, b]$ as follows:

$$a = t_0 < t_1 < t_2 < \cdots < t_{n-1} < t_n = b, \qquad \text{where } n \ge 2.$$

As shown in the above Figure, over a typical subinterval $[t_i, t_{i+1}]$, the distance Δs_i traveled along the curve is approximately $\sqrt{\Delta x_i^2 + \Delta y_i^2}$, by the Pythagorean theorem. Hence, if the subinterval is sufficiently small, the work done in moving the object along this portion of the curve is approximately

$$\text{Force} \times \text{Distance} \approx f(x_{i_*}, y_{i_*})\sqrt{\Delta x_i^2 + \Delta y_i^2},$$

where $(x_{i_*}, y_{i_*}) = (x(t_{i_*}), y(t_{i_*}))$ for some $t_{i_*} \in [t_i, t_{i+1}]$. Therefore, the total work done along the entire curve is approximately

$$W \approx \sum_{i=0}^{n-1} f(x_{i_*}, y_{i_*})\sqrt{\Delta x_i^2 + \Delta y_i^2}.$$

Since

$$\sqrt{\Delta x_i^2 + \Delta y_i^2} = \sqrt{\left(\frac{\Delta x_i}{\Delta t_i}\right)^2 + \left(\frac{\Delta y_i}{\Delta t_i}\right)^2}\,\Delta t_i,$$

where $\Delta t_i = t_{i+1} - t_i$, we obtain

$$W \approx \sum_{i=0}^{n-1} f(x_{i_*}, y_{i_*}) \sqrt{\left(\frac{\Delta x_i}{\Delta t_i}\right)^2 + \left(\frac{\Delta y_i}{\Delta t_i}\right)^2}\, \Delta t_i.$$

Taking the limit as the length of the largest subinterval tends to 0, the sum becomes an integral over $[a, b]$. The difference quotients $\frac{\Delta x_i}{\Delta t_i}$ and $\frac{\Delta y_i}{\Delta t_i}$ converge to $x'(t)$ and $y'(t)$, respectively, while $f(x_{i_*}, y_{i_*})$ approaches $f(x(t), y(t))$. Hence,

$$\boxed{W = \int_a^b f(x(t), y(t)) \sqrt{x'(t)^2 + y'(t)^2}\, dt.}$$

Definition 17. *For a real-valued function $f(x, y)$ and a curve C in $\mathbb{R}^2$, parametrized by $x = x(t)$, $y = y(t)$, $a \le t \le b$, the* ***line integral (of a scalar field)*** *of $f(x, y)$ along C with respect to arc length s is*

$$\int_C f(x, y)\, ds = \int_a^b f(x(t), y(t)) \sqrt{x'(t)^2 + y'(t)^2}\, dt.$$

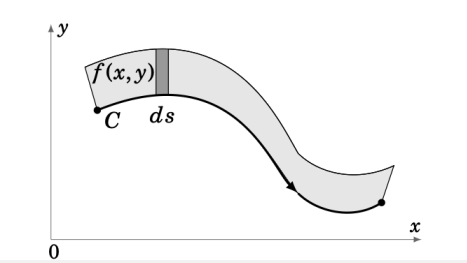
The function $s = s(t)$ gives the arc length of the curve C from $t = a$ to t, for each $t \in [a, b]$. Thus,

$$s = s(t) = \int_a^t \sqrt{x'(u)^2 + y'(u)^2}\, du.$$

By the Fundamental Theorem of Calculus,

$$ds = s'(t)\, dt = \sqrt{x'(t)^2 + y'(t)^2}\, dt.$$

For a general real-valued function $f(x, y)$, what does the line integral $\displaystyle\int_C f(x, y)\, ds$ represent? The preceding discussion of ds gives us a clue. We can think of differentials as infinitesimal lengths. Thus, if we regard $f(x, y)$ as the height of a picket fence along C, then $f(x, y)\, ds$ represents approximately the area of a small section of the fence. Consequently, the line integral $\displaystyle\int_C f(x, y)\, ds$ represents the total area of the fence along the curve C (see the figure below).



Example 41. *Use a line integral to show that the lateral surface area A of a right circular cylinder of radius r and height h is $2\pi rh$.*
Solution: *Let C be the base circle of the cylinder, given by*

$$x^2 + y^2 = r^2.$$

Parametrize C by

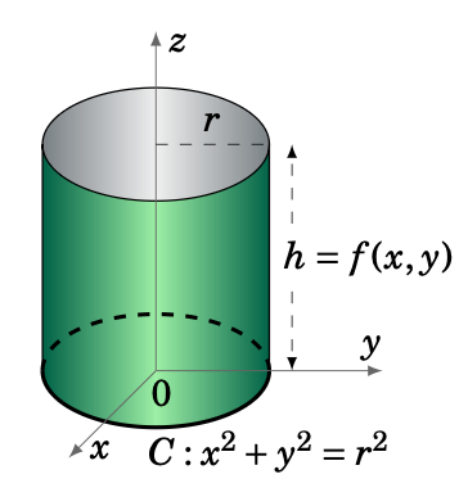
$$x = x(t) = r\cos t, \qquad y = y(t) = r\sin t, \qquad 0 \le t \le 2\pi.$$

Since the height of the cylinder is constant, let $f(x, y) = h$. Then the lateral surface area is

$$\begin{aligned} A &= \int_C f(x,y)\,ds \\ &= \int_0^{2\pi} f(x(t),y(t))\sqrt{x'(t)^2 + y'(t)^2}\,dt \\ &= \int_0^{2\pi} h\sqrt{(-r\sin t)^2 + (r\cos t)^2}\,dt \\ &= hr\int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t}\,dt \\ &= hr\int_0^{2\pi} 1\,dt \\ &= 2\pi rh. \end{aligned}$$

Hence, the lateral surface area of the cylinder is

$$\boxed{A = 2\pi rh}.$$



The force acting on an object is a vector quantity. This motivates us to extend the line integral to vector fields. Let

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j},$$

where P and Q are continuous real-valued functions. For a curve C parametrized by $x = x(t)$ and $y = y(t)$, $a \leq t \leq b$, let

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$$

be its position vector. Then

$$\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}.$$

Using the dot product, we obtain

$$\begin{aligned}\int_C P(x, y)\,dx + \int_C Q(x, y)\,dy &= \int_a^b [P(x(t), y(t))x'(t) + Q(x(t), y(t))y'(t)]\,dt \\ &= \int_a^b \mathbf{F}(x(t), y(t)) \cdot \mathbf{r}'(t)\,dt.\end{aligned}$$

This leads to the following definition.

Definition 18. *For a vector field* $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ *and a curve* C *with a smooth parametrization* $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, *the* ***line integral (of a vector field)*** *of* $\mathbf{F}$ *along* C *is*

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C P(x, y)\,dx + \int_C Q(x, y)\,dy \\ &= \int_a^b \mathbf{F}(x(t), y(t)) \cdot \mathbf{r}'(t)\,dt,\end{aligned}$$

where $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ *is the position vector of points on* C.

Recall that $\mathbf{r}'(t)$ is a tangent vector to the curve C at the point $(x(t), y(t))$, pointing in the direction of increasing t. Since C is smooth, $\mathbf{r}'(t) \neq \mathbf{0}$ on $[a, b]$. Hence,

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

is the unit tangent vector to C at $(x(t), y(t))$. Combining the definitions of the two types of line integrals, we obtain the following theorem.

Theorem 13. *Let $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ be a vector field, and let C be a smooth curve parametrized by $x = x(t)$, $y = y(t)$, $a \leq t \leq b$, with position vector*

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}.$$

Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\, ds,$$

where $\mathbf{T}(t) = \dfrac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ is the unit tangent vector to C.

If the vector field $\mathbf{F}(x, y)$ represents the force acting on an object as it moves along a curve C, then the work W done by this force is

$$W = \int_C \mathbf{F} \cdot \mathbf{T}\, ds = \int_C \mathbf{F} \cdot d\mathbf{r}.$$

Example 42. *Evaluate the line integral $\displaystyle\int_C f(x, y)\, ds$, where*

$$f(x, y) = \frac{x}{x^2 + 1}, \qquad C: \ x = t, \ y = 0, \ 0 \leq t \leq 1.$$

Solution: *The curve C is parametrized by*

$$x(t) = t, \qquad y(t) = 0, \qquad 0 \leq t \leq 1.$$

Using the formula for a line integral with respect to arc length, we obtain

$$\begin{aligned}\int_C f(x,y)\,ds &= \int_0^1 f(x(t),y(t))\sqrt{(x'(t))^2+(y'(t))^2}\,dt\\ &= \int_0^1 \frac{t}{t^2+1}\sqrt{1^2+0^2}\,dt\\ &= \int_0^1 \frac{t}{t^2+1}\,dt\\ &= \frac{1}{2}\left[\ln(t^2+1)\right]_0^1\\ &= \boxed{\frac{\ln 2}{2}}.\end{aligned}$$

Although we defined line integrals over a single smooth curve, we can also define them over a *piecewise smooth curve*. If

$$C = C_1 \cup C_2 \cup \cdots \cup C_n,$$

where $C_1, \ldots, C_n$ are smooth curves, then

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_{C_1} \mathbf{F}\cdot d\mathbf{r}_1 + \int_{C_2} \mathbf{F}\cdot d\mathbf{r}_2 + \cdots + \int_{C_n} \mathbf{F}\cdot d\mathbf{r}_n,$$

where each $\mathbf{r}_i$ is the position vector of the curve C_i.

Example 43. *Evaluate* $\displaystyle\int_C (x^2+y^2)\,dx + 2xy\,dy$, *where* C *is the polygonal path from* $(0,0)$ *to* $(0,2)$ *to* $(1,2)$.
Solution: *Write* $C = C_1 \cup C_2$, *where* C_1 *is given by* $x = 0$, $y = t$, $0 \le t \le 2$, *and* C_2

is given by $x = t$, $y = 2$, $0 \le t \le 1$ *(see Figure 18). Then*

$$\begin{aligned}
&\int_C (x^2 + y^2)\,dx + 2xy\,dy \\
&= \int_{C_1} (x^2 + y^2)\,dx + 2xy\,dy + \int_{C_2} (x^2 + y^2)\,dx + 2xy\,dy \\
&= \int_0^2 \left[(0^2 + t^2)(0) + 2(0)t(1)\right] dt \\
&\quad + \int_0^1 \left[(t^2 + 4)(1) + 2t(2)(0)\right] dt \\
&= \int_0^2 0\,dt + \int_0^1 (t^2 + 4)\,dt \\
&= \left[\frac{t^3}{3} + 4t\right]_0^1 = \frac{1}{3} + 4 = \boxed{\frac{13}{3}}.
\end{aligned}$$

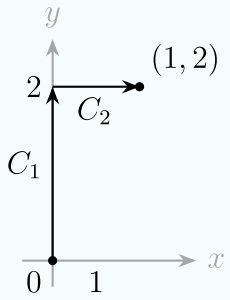


Figure 18: The polygonal path $C = C_1 \cup C_2$.

Example 44. *Show that if* $\mathbf{F} \perp \mathbf{r}'(t)$ *at each point* $\mathbf{r}(t)$ *along a smooth curve* C*, then*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0.$$

Solution: *Let* $\mathbf{r}(t)$, $a \le t \le b$, *be a smooth parametrization of* C*. By the definition of the line integral,*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt.$$

Since $\mathbf{F}$ *is perpendicular to* $\mathbf{r}'(t)$ *at every point along* C*, we have*

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0.$$

Therefore,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b 0\,dt = \boxed{0}.$$

Example 45. *Let C be a smooth curve of arc length L, and suppose that $\mathbf{F}(x,y) = P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}$ is a vector field such that $\|\mathbf{F}(x,y)\| \le M$ for all (x,y) on C. Show that*

$$\left|\int_C \mathbf{F}\cdot d\mathbf{r}\right| \le ML.$$

Solution: *Let $\mathbf{r}(t)$, $a \le t \le b$, be a smooth parametrization of C. By the definition of the line integral,*

$$\begin{aligned}\left|\int_C \mathbf{F}\cdot d\mathbf{r}\right| &= \left|\int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt\right| \\ &\le \int_a^b |\mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)|\,dt \\ &\le \int_a^b \|\mathbf{F}(\mathbf{r}(t))\|\,\|\mathbf{r}'(t)\|\,dt \\ &\le M\int_a^b \|\mathbf{r}'(t)\|\,dt \\ &= ML.\end{aligned}$$

Here, we used the triangle inequality for integrals, the Cauchy–Schwarz inequality, and the fact that

$$L = \int_a^b \|\mathbf{r}'(t)\|\,dt.$$

Hence,

$$\boxed{\left|\int_C \mathbf{F}\cdot d\mathbf{r}\right| \le ML.}$$

4.2 Properties of line integrals

We now discuss some basic properties of line integrals. These properties will simplify the evaluation of line integrals and help us understand how they behave under reversing the direction of curves or under closed curves.

Theorem 14. *Let C be a smooth curve and let $f(x,y)$ be a continuous real-valued function on C. If $-C$ denotes the same curve traversed in the opposite direction, then*

$$\int_C f(x,y)\,ds = \int_{-C} f(x,y)\,ds.$$

Proof. Suppose C has a smooth parametrization $x = x(t)$, $y = y(t)$, $a \le t \le b$. Then

$-C$ is parametrized by

$$x = x(a+b-t), \qquad y = y(a+b-t), \qquad a \le t \le b.$$

Using the formula for a line integral with respect to arc length, we obtain

$$\int_{-C} f(x,y)\,ds = \int_a^b f(x(a+b-t), y(a+b-t)) \cdot \sqrt{[x'(a+b-t)]^2 + [y'(a+b-t)]^2}\,dt.$$

Making the substitution $u = a+b-t$, we get

$$\begin{aligned}\int_{-C} f(x,y)\,ds &= \int_a^b f(x(u), y(u))\sqrt{[x'(u)]^2 + [y'(u)]^2}\,du \\ &= \int_C f(x,y)\,ds.\end{aligned}$$

Therefore,

$$\boxed{\int_C f(x,y)\,ds = \int_{-C} f(x,y)\,ds.}$$

□

For a scalar field, reversing the direction of the curve does not change the value of the line integral $\int_C f\,ds$, whereas for a vector field, reversing the direction of the curve changes the sign of the line integral $\int_C \mathbf{F}\cdot d\mathbf{r}$.

Theorem 15. *Let $\mathbf{F}(x,y) = P(x,y)\mathbf{i} + Q(x,y)\mathbf{j}$ be a continuous vector field, and let C be a smooth curve. If $-C$ denotes the same curve traversed in the opposite direction, then*

$$\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}.$$

Proof. Let C have a smooth parametrization $x = x(t)$, $y = y(t)$, $a \le t \le b$. Then $-C$ is parametrized by

$$x = x(a+b-t), \qquad y = y(a+b-t), \qquad a \le t \le b.$$

Using the definition of the line integral, we obtain

$$\begin{aligned}\int_{-C} P(x,y)\,dx &= \int_a^b P(x(a+b-t), y(a+b-t))\frac{d}{dt}x(a+b-t)\,dt \\ &= -\int_a^b P(x(a+b-t), y(a+b-t))x'(a+b-t)\,dt.\end{aligned}$$

Making the substitution $u = a + b - t$, we get

$$\begin{aligned}\int_{-C} P(x,y)\,dx &= \int_b^a P(x(u),y(u))x'(u)\,du \\ &= -\int_a^b P(x(u),y(u))x'(u)\,du \\ &= -\int_C P(x,y)\,dx.\end{aligned}$$

Similarly,

$$\int_{-C} Q(x,y)\,dy = -\int_C Q(x,y)\,dy.$$

Therefore,

$$\begin{aligned}\int_{-C} \mathbf{F}\cdot d\mathbf{r} &= \int_{-C} P(x,y)\,dx + \int_{-C} Q(x,y)\,dy \\ &= -\int_C P(x,y)\,dx - \int_C Q(x,y)\,dy \\ &= -\int_C \mathbf{F}\cdot d\mathbf{r}.\end{aligned}$$

Hence the result follows. □

The formula

$$\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}$$

has a natural interpretation in terms of work. If a force $\mathbf{F}$ moves an object along C and then back to its initial point along the same path, the total work done is zero. This is because the line integral of a vector field takes the direction of motion into account.